

Complexity Metabolic Failure: A Conceptual Reconstruction of Systemic Limits in the Post-Growth Era

—A Theoretical Construction Based on the Abductive Locking Method

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Theoretical Affiliation

This paper is part of the Civilization Dynamics research program.

Theoretical foundations: *Systematic Ontology* and *Noospheric Ecology*.

(Currently available in Chinese only. English translations will follow.)

Methodological Declaration

This paper employs the Abductive Locking Method as the procedure for concept discovery and verification.

The argumentative structure follows the logical path of "negative delimitation—kernel distillation—boundary disclosure."

A "Concept Preview and Reading Guide" is provided before the main text; readers are advised to consult it first.

In case of interpretive discrepancies, the Chinese original shall prevail.

Abstract

Problematic Contemporary global society exhibits a cluster of systemic predicaments that resist adequate capture by existing conceptual categories: the diminishing marginal returns of *involution*, the predictive failure of *limits to growth*, the recurrence of *systemic risk*, and the synchronous intensification of *governance crises*. These phenomena collectively point toward a deeper constraint that has yet to receive precise conceptual designation—a constraint that is neither classical Malthusian resource scarcity, nor technological stagnation, nor moral decline. The central task of this paper is the conceptual reconstruction and formal naming of this epochal systemic limit.

Method This paper presupposes no prior theoretical position and does not rely on the conventional path of literature review. Concept discovery and verification strictly follow the five-step procedure of the Abductive Locking Method: pattern recognition of boundary crises, logical inversion, delimitation and locking of possibility space, stress testing, and formalized statement. By systematically excluding "what it is not" (an eightfold negative delimitation), the paper approaches the residual kernel that no existing category can capture.

Core Concept Construction The exclusion procedure converges on a formalized definition: when the total complexity generated by the horizontal interactions among the subsystems of a noospheric system persistently exceeds the information-processing and coordination bandwidth ceiling of that system's top-level governance protocol, the system enters an irreversible state of functional congestion. This paper formally designates this state **Complexity Metabolic Failure (CMF)**. Its kernel postulate is:

Protocol Metabolic Capacity < Interactive Complexity Output → Functional Congestion Irreversible

This concept passes the necessity test under the principle of Occam's Razor: it subsumes under a single category the anomalous phenomena that *involution*, *limits to growth*, *systemic risk*, and *governance crisis* each fail to explain independently, achieving a theoretical simplification in which one concept replaces four.

Boundaries and Criteria This paper proactively discloses the concept's boundary of validity: the dynamic state it describes dissolves only under two philosophical limit assumptions—either "the interactive complexity of the system is artificially reduced to zero," or "there exists a Laplace's demon-like coordinator of infinite computational power." Neither condition is instantiable in human civilizational systems, establishing the concept's practical stability. The paper further proposes three operationalizable draft criteria—(A) the growth rate of macro-coordination costs exceeds that of utility; (B) the half-life of crisis incubation is shorter than the response cycle of rule revision; (C) the explanatory coverage rate of the meta-narrative undergoes irreversible decline—thereby laying tracks for subsequent quantitative testing.

Theoretical Implications The introduction of Complexity Metabolic Failure accomplishes a structural turn in the Malthusian–Limits-to-Growth theoretical lineage: it pivots the unit of analysis for systemic limits from the inflow end of material resources (food, energy, raw materials) to the metabolic end of information and protocol (coordination capacity, cognitive bandwidth, meta-narrative explanatory power). Within this framework, the essence of the contemporary global predicament is revealed as the critical pangs of civilization's phase transition from *collective mind* to *synergistic cognitive network*—the old meta-narrative has exhausted its protocol metabolic surplus, while a new civilization-level cognitive architecture has yet to take form.

Keywords: Complexity Metabolic Failure; Systemic Limits; Abductive Locking Method; Protocol Metabolic Capacity; Meta-Narrative; Concept Construction; Civilizational Phase Transition

Concept Preview and Reading Guide

The argumentative structure of this paper differs from that of a conventional academic article in three notable respects. The following notes are intended to help the reader establish appropriate expectations and become familiar in advance with the paper's core terminological system.

I. Overview of Original Concepts Introduced in This Paper

The following concepts are proposed and systematically constructed for the first time in this paper. The parentheses contain condensed versions of their operational definitions; the main text will unfold them layer by layer.

Concept	Condensed Definition
Complexity Metabolic Failure	A state of irreversible functional congestion into which a system falls when the total complexity generated by horizontal subsystem interactions persistently exceeds the information-processing bandwidth of the top-level governance protocol.
Protocol Metabolic Capacity	The upper limit of information-processing capacity by which the system's top-level governance architecture can, per unit time, effectively interpret, coordinate, and respond to subsystem interaction events.
Interactive Complexity Output	The total incremental rate of novel interactive relations, unintended consequences, and coordination demands generated among subsystems within a system as a result of self-organizing evolution.
Functional Congestion	A dynamic state in which the system's overall efficacy, owing to exhausted coordination bandwidth, exhibits valid input but disordered output—distinct from collapse: the system continues to operate, yet each additional unit of input yields diminishing or even negative net returns.

II. Existing Concepts Introduced and Re-Operationalized in This Paper

The following concepts originate from the author's published theoretical systems, Systematic Ontology and Noospheric Ecology. This paper does not presuppose reader familiarity with those systems. Each concept, upon first use, is given the minimal operational definition required for this paper's purposes, either in a footnote or in an accompanying parenthetical note.

Concept	Minimal Operational Definition Used in This Paper	Theoretical Source
Meta-Narrative	The highest-order shared cognitive framework of a civilization or social system. It does not directly participate in the material operations of the system, but defines "what is legitimate," "what is possible," and "in which direction the system should evolve." [1]	<i>Noospheric Ecology</i> , Corollary to Axiom IV
Synergistic Cognitive Network	An idealized state of systemic architecture in which the horizontal interactions of subsystems can be perceived and coordinated in real time through a metacognitive interface, endowing the system as a whole with the capacity to observe and regulate its own operation.	<i>Noospheric Ecology</i> , Theorem A0
Antagonistic Failure	The phenomenon in which antagonistic forces originally possessing corrective functions within a system (such as public opinion oversight, competitive mechanisms, and checks-and-balances structures) lose their capacity for effective feedback due to information overload or protocol closure.	<i>Noospheric Ecology</i> , Corollary to Theorem D22.1
Abductive Locking Method	A procedure for concept discovery and verification: by exhausting all logically possible positions and eliminating them one by one using incoherence or explanatory redundancy as criteria, it ultimately approaches the residual kernel that survives stress testing. [2]	<i>Systematic Ontology</i> , Appendix to Introduction
Noospheric System	A complex system with information processing at its core, meaning construction as its driving force, and symbols as its hereditary units. Civilizations, organizations, and individual minds all belong to this category.	<i>Noospheric Ecology</i> , Axiom II

[1] In this paper, "Meta-Narrative" is defined as a topological constraint on the system's possibility space—not to be conflated with Lyotard's postmodern critique of *grands récits*. The concept is operationalized here within the framework of systems ontology, not post-structuralist critical theory.

[2] For a complete epistemological defense of the Abductive Locking Method, see He (2024), 溯因封锁法 / *The Abductive Locking Method*, Zenodo. <https://doi.org/10.5281/zenodo.19663968>. The English translation is provided for accessibility; the Chinese original remains the authoritative text for any interpretive disputes.

III. Special Notes on Style and Structure

(1) Why does this paper lack a conventional "Literature Review" section?

A conventional academic paper uses a front-loaded literature review to anchor itself within a knowledge lineage and identify a "research gap." This paper did not grow out of the internal interstices

of the existing literature; it emerged from the pattern recognition of boundary crises across disciplines—its starting point is a phenomenon that has yet to be named, not a question that has already been thoroughly discussed.

Embedding a literature review before the main text would create a misalignment in the argumentative path: the reader would be guided to read this paper with the expectation of "filling a gap," whereas what this paper actually performs is "concept midwifery"—bestowing a precise categorical name upon a syndrome long misdiagnosed.

For this reason, the paper places its dialogue with existing theories in Chapter 6, "Dialogue and Delimitation." This is a logically consequential treatment: first establish the new concept, then demarcate its boundaries from old categories. Readers accustomed to the conventional structure may preview Chapter 6 to establish their coordinates before returning to Chapter 4 for the conceptual kernel.

(2) Does the argument of this paper depend on prior theories such as *Systematic Ontology*?

Not entirely. The argument of this paper adopts a principle of "minimal self-sufficiency": you do not need prior familiarity with *Systematic Ontology* or *Noospheric Ecology* to read and evaluate it. All concepts drawn from the parent theoretical systems have been given paper-specific operational definitions in the table above.

The theoretical lineage traced in the footnotes serves as a "deep reading interface" for interested readers—it indicates the coordinates of this paper within a larger theoretical landscape, but does not constitute a necessary premise for the argument. Skipping all footnotes will not impair comprehension of the core argument.

(3) Why is the density of new concepts so high? Does this constitute "academic jargon"?

This paper does introduce a certain density of new terminology. The necessity of this quantity arises from the structural nature of the concept itself: to precisely describe an entirely new type of systemic limit, one must simultaneously equip it with a set of companion terms describing its dynamic elements (complexity output at the input end, metabolic bandwidth at the output end, congestion at the failure end). Forcing old words to do this work would risk even graver misunderstanding through semantic wear.

To prevent these concepts from degenerating into jargon, the paper adopts three defensive measures:

- **The Concept Preview layer** (i.e., this section): the reader holds the full map before entering the main text.
- **Definitions accompanying the text inline**: each definition, upon first use in the main text, is given an operational explication in a standalone paragraph or bracketed gloss.
- **Isomorphic model as aid**: Chapter 4 uses "Windows XP's failure to support new applications" as an intuitive model, helping the reader grasp abstract concepts within a concrete scenario.

(4) Suggested Reading Paths

- **For quick comprehension**: Read Abstract → Concept Preview (this section) → Chapter 4 (Conceptual Kernel) → Chapter 8 (Conclusion). This path allows one to grasp the core insight within 15 minutes.

- **For the complete argument:** Read sequentially. Chapters 2 and 3, on methodology and the exclusion procedure, are the legal foundation of the concept; skipping them is not recommended.
- **For academic dialogue:** Prioritize Chapter 6 (Theoretical Delimitation) and Chapter 7 (Implications and Roadmaps), then return to the core argument.

Chapter 1 Introduction

1.1 Symptoms: An Era That Cannot Be Captured by the Old Dictionary

Human civilization in the 2020s is experiencing a peculiar kind of aphasia.

We have an ample vocabulary for describing local crises—"climate change," "supply chain disruption," "democratic backsliding," "meaning crisis," "fertility collapse," "AI ethics dilemmas"—but when we try to articulate the common grammar underlying these crises, our lexicon abruptly goes blank.

This is not a predicament of rhetoric. The impotence of language often signals that a cognitive framework has hit its ceiling.

Consider the following three sets of juxtaposed phenomena. Each cuts across the boundaries of traditional disciplines.

Group One: Efficiency Boomeranging

The sophistication of global supply chains has reached a historical peak—a stoppage at a single port in Shenzhen can shut down a factory in Detroit within 72 hours. Trading latency in financial markets has been compressed to microseconds. The granularity of corporate management now reaches down to the keystroke frequency of individual employees. And yet we simultaneously observe: a micro-perturbation at any node of the supply chain propagates through the entire network with amplified amplitude; the frequency of financial market "flash crashes" has increased by nearly two orders of magnitude over the past two decades; workplace burnout and the culture of "lying flat" have spread from fringe subcultures to global phenomena.

These are not isolated "management problems" or "market failures." They share a single dynamic signature: the complexity that a system adds to improve local efficiency has, in turn, become the root cause of its own systemic fragility.

Group Two: When Remedies Compound into Poison

Policies, each perfectly reasonable on its own, produce absurd outcomes when superimposed.

Biofuel subsidies introduced to curb climate change drove up global food prices and triggered food riots in multiple countries in 2008. Regulatory compliance requirements strengthened to prevent financial risk have crushed small and medium-sized banks, ultimately concentrating market share further into "too big to fail" institutions. Trade barriers erected to protect domestic employment have disrupted intermediate goods trade, thereby weakening the competitiveness of domestic manufacturing itself.

These are not cases of "poor policy design." They share a single logical predicament: each subsystem—environmental protection, finance, trade—is acting according to its own optimal strategy, but the interaction effects among subsystems exceed the information-processing capacity of any single decision-

maker.

Group Three: The Shortening Half-Life of Trust Narratives

Meta-narratives that once ran steadily for decades or even centuries—the fundamental stories defining "who we are" and "where we are headed"—are visibly eroding.

The intergenerational contract that "hard work will improve your life" has collapsed under the weight of housing-price-to-income ratios and credential inflation. The globalization narrative of "free trade brings universal prosperity" has crumbled before rust-belt factories and decoupling rhetoric. The Enlightenment narrative that "democracy selects the best governance" has been hollowed out by political polarization and populist waves. Even the techno-optimism of "humanity will colonize Mars" has betrayed a certain absent-minded exhaustion amid recent global turmoil.

These are not mere "ideological contests" or "crises of faith." They share a single structural signal: the internal complexity of the old narratives can no longer accommodate the genuinely novel experiences generated by subsystem evolution. The stories have not yet been refuted, but they can no longer "contain" reality.

1.2 Diagnosis: The Collective Inaccuracy of Old Categories

When the above phenomena are submitted to public discourse, four sets of diagnostic categories are promptly activated. But each activation is accompanied by a leakage of explanatory power.

"This is involution."

Involution describes diminishing marginal returns within a single dimension—more people competing for the same pie, each person getting less. But it cannot explain why systemic anxiety and disorder are intensifying even in domains where technology continues to advance and the pie is still growing (such as the internet platform economy). Involution presupposes closed boundaries, yet the contemporary predicament is unfolding precisely at the intersections of cross-dimensional interactions—the progress of AI has not made programmers "more involuted"; it has redefined the very ecological niche of "programming" itself. Involution is inward collapse; what we are living through is outward bursting.

"This is the limits to growth."

The Limits to Growth paradigm of the Club of Rome describes physical ceilings on the input side—resource depletion, energy shortages, environmental carrying capacity. This paradigm has profoundly shaped half a century of environmentalism. Yet applying it to diagnose the present moment encounters an awkward fact: we are not living in material scarcity. Global food output is sufficient to feed everyone; the cost of clean energy has fallen below that of fossil fuels; the technology supply side has not stalled. The predicament lies not in "what is flowing in," but in "where the outflow goes and how it is metabolized." Carbon emissions are not a problem of resource scarcity—they are a problem of metabolism. Plastic pollution is not a shortage of raw materials—it is a problem of metabolism. Limits

to growth is suffocation at the inflow end; what we are living through is congestion at the outflow end.

"This is systemic risk."

Systemic risk describes the cascading collapse produced when local failures propagate through a tightly coupled network—"too big to fail" is its classic formulation. But the crisis transmission mechanism it describes is event-triggered: a Lehman Brothers collapses, and the contagion then spreads through the network. In the contemporary predicament, we observe something more uncanny: the absence of a clear triggering event, yet the system is nevertheless spontaneously losing coherence. No single bank has failed, yet the financial system is permeated by a liquidity illusion. No country has declared war, yet globalization is quietly unraveling. No authority has proclaimed "truth is dead," yet the riverbed of consensus is drying up imperceptibly. Systemic risk explains "fuse and explosion"; what we face is a sustained smoldering without a fuse.

"This is a governance crisis."

Governance crisis attributes disorder to the failure of political institutions or managerial capacity—corruption, short-sightedness, elite capture, poor execution. But it cannot explain a deeper paradox: why do even the most well-intentioned policies produce synthetic fallacies when superimposed? Why do governance systems with the highest-quality bureaucracies, the most advanced information systems, and the most transparent decision-making procedures also stumble over climate, migration, and inequality? If only "bad governance" fails, we could fix things by replacing the governors; but if even "good governance" is failing, then what is failing is not the people, but the very architecture they operate within.

Four categories, each capturing one facet of the symptoms. But piecing them together, we still do not obtain a complete diagnosis—because the fundamental category that would unify these symptoms into a single syndrome has yet to be named.

1.3 The Task: A Rigorous Concept Midwifery

The task this paper undertakes is not to add a fifth "perspective" alongside the four categories above. Adding perspectives only makes our lexicon more crowded without curing the aphasia.

The task of this paper is a conceptual level-shift: to ascend from descriptive categories that are valid only within specific subsystems to a constructive category capable of characterizing the dynamic state of the system as a whole.

Specifically, this paper performs three tasks.

First, it establishes a unified dynamic diagnosis for the contemporary global predicament. We will argue that involution, the failure of limits to growth, the smoldering form of systemic risk, and the architectural roots of governance crisis—these apparently heterogeneous phenomena are differential manifestations of a single underlying state. That state is this: the total complexity generated by the horizontal interactions among subsystems within the system has persistently exceeded the information-processing bandwidth

ceiling of the top-level governance protocol. We name this state Complexity Metabolic Failure.

Second, it establishes the legitimacy of the concept through a rigorous abductive procedure. This paper does not appeal to "insight" or "intuition" to arrive at this concept. The discovery and verification of the concept follow a publicly testable procedure—the Abductive Locking Method. We will demonstrate how this category was approached as "the only one left standing" through the exhaustion and sequential elimination of the logically possible positions. It is not a clever metaphor; it is a theoretical entity tempered by methodological rigor.

Third, it lays down interfaces for subsequent quantitative and design-oriented research. Concept construction is the starting point of a theoretical chain, not its terminus. While establishing the conceptual kernel, this paper will simultaneously specify its boundary of validity (under what assumptions it loses explanatory power) and its draft criteria (how to observe its occurrence and measure its severity). These will constitute the tracks leading from "qualitative diagnosis" toward "quantitative testing" and "architectural design."

1.4 Preliminary Notes on Conceptual Naming

Before entering the formal argument, it is necessary to offer some preliminary etymological clarification of the paper's core term—Complexity Metabolic Failure.

Complexity. The "complexity" used in this paper is not the everyday-language sense of "complicatedness" or "trouble." It designates specifically the information increment generated by the horizontal interactions among subsystems within a system—not the number of subsystems themselves, but the novel interactive relations, unintended consequences, and coordination demands that these subsystems continuously produce through their self-organizing evolution. A system with only ten subsystems that are tightly coupled to one another has far higher complexity than a system with a hundred subsystems that are mutually isolated.

Metabolic. This term is borrowed from biology, but is redefined in this paper as the process of information processing and coordination performed by the system's top-level governance protocol upon the interaction effects of subsystems. It comprises three links: (a) perception—whether an interaction event can be effectively registered; (b) interpretation—whether the causal structure of the interaction event can be correctly attributed and understood; (c) response—whether effective coordinating actions can be generated and executed. A bandwidth insufficiency in any one of these three links constitutes a metabolic bottleneck.

Failure. This paper uses "failure" rather than "collapse" or "crisis" in order to precisely distinguish between two dynamic states. Collapse means that the system ceases to operate or disintegrates; failure means that the system continues to operate, but has entered functional congestion—input remains valid (resources, technology, rules continue to be injected), but output is disordered (each additional unit of input yields diminishing or even negative net returns). Failure is the precursor to collapse, but it is ontologically an independent dynamic phase.

The composite meaning of Complexity Metabolic Failure is: when the information increment (complexity) produced by horizontal subsystem interactions persistently exceeds the total bandwidth (metabolic capacity) of the top-level governance protocol across its three links of perception, interpretation, and response, the system enters a state of functional congestion (failure) in which input remains valid but output is disordered. The kernel postulate of this concept will be given formal statement in Chapter 4.

1.5 Roadmap of the Paper

Chapter 2 presents the methodological instrument of this paper—the five steps of the Abductive Locking Method—and declares the theoretical anchor points of the paper. This chapter is the foundation of the concept's legitimacy; skipping it is not recommended.

Chapter 3 executes the exclusion procedure. We systematically argue "what the contemporary global predicament is not": not a classical Malthusian trap, not energy depletion, not technological stagnation, not absolute overpopulation, not a decline in civilizational will, not a single-dimensional crisis, not an ordinary cyclical trough, not the moral or intellectual failure of any particular group. After the eightfold exclusion, the residual contour will naturally emerge.

Chapter 4 is the theoretical heart of the paper. We provide the formalized definition and kernel postulate of Complexity Metabolic Failure, aided by an isomorphic model from computer systems. We then test the concept's necessity under Occam's Razor: why replacing four old terms with this single term is the theoretically superior choice.

Chapter 5 proactively discloses the boundaries and criteria of the concept. We specify under which philosophical limit assumptions the concept would cease to hold, and why these assumptions are not instantiable in human civilization. We then propose three operationalizable draft criteria, laying tracks for the empirical testing of the concept.

Chapter 6 executes a dialogue and delimitation with existing paradigms. We systematically compare Complexity Metabolic Failure with the Malthusian trap, the Limits to Growth thesis, the concept of involution, and systemic risk theory, thereby clarifying the ecological niche of the present concept within the tree of human knowledge.

Chapter 7 sets out the theoretical implications and roadmaps for subsequent research. We explain how Complexity Metabolic Failure, as a diagnostic framework, points toward the Synergistic Cognitive Network as a constructive counterpart. We also clarify the boundary of this paper: it is a diagnostic instrument, not a prescription pad.

Chapter 8, the Conclusion, restates the core findings and responds to the ultimate question posed by this concept: is it a fatalistic death sentence, or the starting point for civilization's self-awareness?

We now begin with methodology.

Chapter 2: Methodology: Abductive Locking and Negative Delimitation

2.1 A Prefatory Question: Where Does a Concept Obtain Its Legitimacy?

Any academic work that attempts to propose a new concept must first answer a meta-question: by virtue of what is this concept to be accepted as knowledge, rather than filed away as "yet another neologism"? In normal science, the legitimacy of a new concept typically derives from two sources. Path One: it grows out of the internal interstices of the existing literature—the researcher discovers that existing categories are incompletely classified or insufficiently explanatory, and proposes a supplementary concept to cover the "leftover cases." Path Two: it grows out of newly discovered empirical materials—the researcher observes phenomena that existing theories cannot accommodate, and proposes a theoretical concept to unify these anomalies.

This paper cannot follow either path.

Path One is infeasible because what this paper confronts is not a problem "insufficiently explained" within the framework of the existing literature, but rather the failure of the entire literature framework itself. When the combined set of concepts—*involution*, *limits to growth*, *systemic risk*, *governance crisis*—still cannot provide a complete diagnosis of the current symptoms, what we need is not to patch another word into their interstices. What we need is to shift the level of diagnosis—from concepts that describe symptoms to a constructive category that subsumes these symptoms. This is not a repair job internal to the literature; it is a reconstruction beyond the literature's boundary.

Path Two is also infeasible, because the phenomena this paper confronts are not lacking in empirical materials. Quite the contrary—there is an overabundance of empirical materials: supply chain data, financial market volatility indices, trust decay curves in social surveys, cost-benefit analyses of climate policies. But each of these spans different disciplinary domains, and no single discipline's empirical paradigm can contain them all at once. If Complexity Metabolic Failure is submitted to economics, it will be demanded that it be tested with GDP data; if submitted to political science, with governance indicators; if submitted to ecology, with carbon flux measurements. The empirical toolbox of each discipline can capture one facet of it, but no discipline can capture its totality.

This paper therefore cannot travel the conventional road. We need a third source of legitimacy.

That source is this: a publicly testable procedure for concept generation and verification. If we can demonstrate that this term is not a flash of metaphorical insight, but the sole residual option "locked in" from a possibility space through a rigorous procedure, then its legitimacy derives not merely from its correspondence to literature or data, but from the logical necessity of the procedure itself.

That procedure is the Abductive Locking Method.

2.2 The Abductive Locking Method: A Five-Step Procedure

The Abductive Locking Method[1] is a concept discovery and verification procedure established in the Introduction to *Systematic Ontology*. It is not a proprietary tool of any single discipline, but a meta-methodology—it does not stipulate what counts as "good evidence," only what counts as "good

procedure."

The name reveals its two core operations. "Abductive": working backward from observable systemic behavior, reverse-infer the deeper mechanisms that sustain that behavior. "Locking": by exhausting all logically possible positions on a question and eliminating them one by one using incoherence or explanatory redundancy as criteria, ultimately approach the residual kernel that cannot be eliminated.

The full procedure consists of five steps.

Step One: Pattern Recognition of Boundary Crises

Collect cases of boundary predicaments across disciplines. "Boundary predicaments" refer to anomalous phenomena that arise when systems approach their operational limits and that cannot be assimilated within the original paradigm—Gödelian incompleteness in mathematics, the ultraviolet catastrophe and divergence difficulties in physics, immune self-reference in biology, the paradox of control in social systems. Strip away the discipline-specific content and extract the formally isomorphic dynamic skeleton. The goal of pattern recognition is not to accumulate cases, but to discover the identical structure that recurs at the boundaries of different domains.

Step Two: Logical Inversion

Traditional accounts of boundary predicaments habitually ask "Why do systems fail at their boundaries?" This mode of questioning embeds an unexamined presupposition: that a system's "normal state" is an unproblematic background requiring no explanation, and that only "failure" needs to be diagnosed. The Abductive Locking Method executes a crucial inversion: turn from asking "Why does the system fail?" to asking "How is it that the system has managed to persist stably most of the time?" This inversion transforms "stability" from an assumed background into an achievement that requires active explanation, thereby directing attention toward those deeper, typically invisible constraint mechanisms that sustain stability. It is the functional attenuation of exactly these mechanisms that accounts for the boundary breakdown we ultimately observe.

Step Three: Delimitation and Locking of Possibility Space

All logically possible answer-types to the target question (for example, "What is the fundamental nature of the contemporary global predicament?") are enumerated, forming an exhaustive possibility space. The threshold for "logically possible" is low—as long as a position is not internally contradictory, it enters the space. These positions are then tested one by one against two locking criteria: (1) incoherence—does the position contain internal logical contradictions; (2) explanatory redundancy—can the explanatory power of the position be covered by a more parsimonious position, or does its core claim rely on non-operationalizable metaphysical entities? The standard of elimination is not "falsehood," but "incoherence" or "unnecessary complexity." After multiple rounds of locking, the set of positions remaining in the space constitutes "the most coherent, most parsimonious, and most explanatorily powerful candidates within the current cognitive framework."

Step Four: Stress Testing

The candidate positions that survive the locking procedure are subjected to the harshest boundary cases available for testing. Stress testing includes three types of situations: the position's own limit scenario (pushing its core proposition to its extreme and checking whether it self-destructs); anomalous

phenomena that appear to constitute counterexamples (whether there exist observational facts that the position cannot accommodate); and compatibility with other established regularities (whether it conflicts with theories that have passed long-term testing). If it passes, it is confirmed; if it fails, return to Step Three and check whether the enumeration of the possibility space missed any positions or whether the locking criteria were unfairly applied.

Step Five: Formalized Statement

The final position that survives stress testing is formally stated as an axiom or theorem. The statement structure must include the following elements: dependencies (the more fundamental propositions on which the statement rests), the core proposition (the formalized statement itself), elaboration (an unfolding explication of the proposition), interdisciplinary anchors (correspondence with core problems in relevant disciplines), and boundary criteria (under what conditions the proposition ceases to hold).

The Abductive Locking Method is itself neither an axiom nor a theorem. It is the meta-procedure for discovering axioms and theorems. It does not claim to be the only method of discovery, but it does claim: the propositions obtained through this method are, given the delimited possibility space and criteria, the optimal candidates within the current cognitive framework. Its openness lies in this: if someone can point out an omission in the possibility space, an unfairness in the criteria, or an oversight in the stress testing, the lock will be reopened. This constitutes its mechanism of self-correction.

2.3 The Methodological Execution Route of This Paper

Applying the Abductive Locking Method to the question of this paper—"What is the fundamental nature of the contemporary global predicament?"—we construct the following execution route.

Stage One (corresponding to Chapter 3): Delimitation and Locking of Possibility Space

We enumerate all logically possible diagnostic types for "what the contemporary global predicament might be." This possibility space includes, but is not limited to: the classical Malthusian trap, an energy-depletion-type crisis, technological stagnation theory, overpopulation theory, civilizational will decline theory, single-dimensional crisis theory, cyclical trough theory, conspiracy/moral failure theories. For each, we eliminate it not on the ground of "insufficient evidence," but on the ground of "incoherence" or "explanatory redundancy." Explanatory redundancy is here operationalized as follows: if the set of phenomena explainable by a given diagnosis A is entirely contained within the set of phenomena explainable by a more parsimonious diagnosis B, then A is redundant.

The goal of this stage is to approach, through the exclusion of "what it is not," a residual contour. This residual contour is not yet the concept; it is the set of feature constraints that the concept must satisfy.

Stage Two (corresponding to Chapter 4): Distilling the Conceptual Kernel from the Residual Contour

The set of feature constraints left standing after Stage One constitutes a series of "functional requirements" for the new concept—it must be able to explain those macro-disordering phenomena that are neither resource scarcity, nor technological stagnation, nor moral failure. The task of concept construction is to find the most parsimonious proposition that can simultaneously subsume all feature constraints, without introducing unnecessary theoretical entities.

This stage will produce two outputs: (1) a formalized definition—using controlled vocabulary to delineate the concept's intension and extension; (2) a kernel postulate—reducing the concept to a structural statement that cannot be further reduced, approaching something close to a tautology. The power of the kernel postulate lies in this: to refute it requires denying certain more fundamental conditions of existence (such as causality, or information-processing latency). This constitutes the concept's "logical hardness."

Stage Three (corresponding to Chapter 5): Proactive Disclosure of Boundaries and Criteria

The final act of concept construction is not to argue for the concept's correctness, but to proactively expose its boundaries of failure. What we ask is not "Under what conditions does it hold?" but "Under what conditions would it cease to hold?"

A mature concept should have the courage to declare: under certain limit conditions (which may be non-instantiable in practice), the dynamic it describes will dissolve or deform. This declaration is not a theoretical weakness, but a theoretical self-circumscription—it prevents the concept from being abused beyond its valid range, and it also reserves clear triggering criteria for possible future conceptual revision.

Simultaneously, we provide draft operational criteria for the concept. The criteria need not be immediately quantifiable, but they must specify "If this concept holds, what type of signal should we observe in the data?" Criteria are the track interfaces leading from a qualitative concept to quantitative testing.

2.4 A Defense of the Methodological Choice: Why This Paper Lacks a Conventional Literature Review

Having laid out the methodology of the Abductive Locking Method, we can now squarely address a structural question: why this paper does not include a conventional literature review chapter after the introduction.

A conventional academic paper includes a literature review to perform three functions: first, to display the author's command of existing research; second, to identify a "research gap"; third, to provide legitimacy for the paper's research question—it responds to a question that the existing literature has failed to adequately answer.

In the problem domain of this paper, none of these three functions can be fulfilled by a literature review.

Displaying command of existing research? The argument of this paper spans at least five disciplinary domains—economics (limits to growth and involution), political science (governance crisis), complex systems science (systemic risk), ecology (metabolism and limits), sociology (trust and meta-narrative). To produce a qualified literature review for each would require a standalone disciplinary review paper. Five such reviews stitched together would not be a paper, but a handbook.

Identifying a research gap? The concept this paper seeks to construct—Complexity Metabolic Failure—does not exist in the existing literature. It is not a question that has already been partially discussed and awaits "filling in"; it is a "new entity" that has never before been proposed as an independent category.

When a concept is not yet on the literature map, a literature review cannot tell you its location—you need to draw it first, and then retrospectively annotate its coordinates relative to known landmarks.

Deriving legitimacy for the research question from the literature? The question of this paper was not read out of the literature. It emerged from pattern recognition of boundary crises across disciplines. Its source of legitimacy is not that "predecessors failed to answer it well," but that "predecessors each answered a part, while the category that would subsume all those parts has yet to be named." To argue for this legitimacy, what is needed is not a literature review, but a transparent display of the abductive procedure.

For these reasons, this paper adopts an alternative path: it places its dialogue with existing theories in Chapter 6, "Dialogue and Delimitation."

The difference between dialogue and review is this: a review is a front-loaded source of legitimacy; a dialogue is a back-loaded boundary demarcation. The former says "where predecessors have arrived and where I take over"; the latter says "I have erected this new entity; now let me tell you its relation to those things you already know."

Readers concerned with the academic lineage of this paper may choose to read Chapter 6 before returning to Chapter 4. The argument sequence in the main text is not the sole reading sequence. This elastic design is itself a manifestation of methodological self-awareness.

2.5 A Minimalist Declaration of Theoretical Anchoring

Finally, an unavoidable question: what is the relationship between this paper and the author's published theoretical systems—*Systematic Ontology*, *Noospheric Ecology*, and *Civilization Dynamics*?

The answer has two layers.

First layer: the position of this paper within that system.

This paper is a concept-construction piece within the *Civilization Dynamics* research program. In the division of labor among the theoretical trilogy, *Systematic Ontology* provides meta-axioms at the ontological level, *Noospheric Ecology* provides general theorems for cognitive and social systems, and *Civilization Dynamics* provides diagnostic and design frameworks at the civilizational scale. The work of this paper belongs to the third layer—it provides a foundational diagnostic category for civilization dynamics.

Second layer: the self-sufficiency of the argument of this paper.

Although this paper draws its methodology and certain theoretical terms from the above systems, the argumentative structure of this paper does not presuppose reader familiarity with the parent systems.

Specifically:

- The methodology (the Abductive Locking Method) is given a complete statement independent of the parent systems in Section 2.2. The reader need not consult *Systematic Ontology* to understand and evaluate this procedure.
- The core terms (Meta-Narrative, Synergistic Cognitive Network, Antagonistic Failure) are

given paper-specific minimal operational definitions in the "Concept Preview and Reading Guide." These definitions are not excerpts from the parent systems, but independently formulated versions reconstructed for the needs of this paper's argument.

- The litmus test of self-sufficiency: can a reader who has never heard of "Systematic Ontology" rely solely on the argument of Chapters 2 through 5 of this paper to independently judge the logical quality of the concept "Complexity Metabolic Failure"? The design goal of this paper is: yes.

The declaration of theoretical anchoring is a "deep reading interface" prepared for those readers who wish to trace the deeper roots of the concept. It is not a necessary premise on the argument chain; it is an optional door—push it open, and you will see a larger theoretical landscape; leave it closed, and the argument of this paper remains complete.

Chapter Summary

Chapter 2 has laid out the methodological foundation of this paper. The core procedure of the Abductive Locking Method—pattern recognition of boundary crises, logical inversion, locking of possibility space, stress testing, and formalized statement—constitutes the operational basis of the concept's legitimacy. Section 2.3 maps this procedure onto the structure of the paper: Chapter 3 executes the locking, Chapter 4 distills the kernel, Chapter 5 discloses the boundaries. Section 2.4 explains why a conventional literature review is infeasible in this problem domain, and the logical justification for replacing "review front-loading" with "dialogue back-loading." Section 2.5 completes the minimalist declaration of theoretical anchoring—the argument of this paper is self-sufficient; tracing the parent systems is an optional interface, not a necessary premise.

The methodological groundwork is now complete. We proceed to the locking procedure.

[1] For a complete epistemological defense of the Abductive Locking Method, see He (2024), 溯因封锁法 / *The Abductive Locking Method*, Zenodo. <https://doi.org/10.5281/zenodo.19663968>. The English translation is provided for accessibility; the Chinese original remains the authoritative text for any interpretive disputes.

Chapter 3: Exclusion: What the New Limit Is Not

3.1 Methodological Principles Governing the Exclusion Procedure

Before executing the exclusion procedure itself, three operational principles must be established. They constitute the "procedural justice" foundation of this chapter's argument.

Principle One: The Dual Standard of Locking Criteria

A candidate diagnosis is eliminated only if it satisfies at least one of the following two criteria:

Incoherence: The diagnosis contains internal logical contradictions, or the premises on which it depends conflict with its own inferences. For example, if a predicament is attributed to "absolute resource scarcity," but the observational data show an absolute surplus of resources while the predicament continues to intensify, the diagnosis is incoherent.

Explanatory Redundancy: The set of phenomena that the diagnosis can explain is entirely contained

within the set of phenomena explainable by a more parsimonious diagnosis, and the diagnosis in question additionally requires the introduction of extra theoretical entities or assumptions. For example, if everything that "cyclical downturn" can explain is already covered by the explanatory range of "metabolic failure," while the latter can further explain the smoldering-type disorder that the former cannot accommodate, then the former is redundant.

A diagnosis that meets neither of these two criteria is not thereby "proven true"—it is merely left standing in the possibility space for now, awaiting further screening by stress testing. Locking is not a final verdict; it is a procedure for shrinking an infinite possibility space to a manageable candidate set.

Principle Two: The Unitary Object of Explanation

All the diagnoses excluded in this chapter share one and the same explanatory target—the "three groups of symptoms" exhibited in Section 1.1 of this paper, together with their common features. These symptoms include, but are not limited to: efficiency boomeranging (supply chain fragilization, the increasing frequency of financial flash crashes), the compounding of remedies into poison (the superimposed failure of well-intentioned policies), and the shortening half-life of trust narratives (the decaying explanatory power of meta-narratives).

We cannot permit one diagnosis to say "I explain symptom A" while another says "I explain symptom B," and then declare that the two complement each other. What we need is the category that can subsume A, B, and C as differential manifestations of a single dynamic process. If a diagnosis can cover only a subset of the symptoms and remains mute on the rest, it is not what we are looking for.

Principle Three: The Requirement of Exhaustiveness

The eight exclusions presented in this chapter are not an arbitrary enumeration. They are designed to cover all the main types of position that have been advanced in contemporary public discourse and academic research regarding "the fundamental nature of the current global predicament." Should a reader believe that a ninth type of position exists that has not been excluded, the locking procedure of Chapter 3 must be reopened—this is precisely the self-corrective mechanism of the Abductive Locking Method. At the end of this chapter, I will reserve a "Self-Checklist of Residual Possibilities," explicitly inviting the reader to participate in the refinement of the locking procedure.

3.2 The Eightfold Exclusion

Exclusion One: It Is Not the Classical Malthusian Trap

The diagnosis to be excluded: The essence of the contemporary global predicament is that population growth has outpaced food supply capacity, and the system will forcibly restore equilibrium through famine, war, or pestilence.

Grounds for exclusion: Incoherence.

The dynamic premise of the classical Malthusian trap is this: the growth rate of food output possesses a rigid ceiling determined by the diminishing marginal returns of land, while the growth rate of population, under unconstrained conditions, can persistently surpass that ceiling. In this model, the "limit" is located on the input side—the biophysical productive capacity of the land.

This diagnosis encounters three incoherencies when applied to contemporary symptoms.

First, the ratio of global total food output to total population has risen continuously over the past half-century. FAO data indicate that per capita global food supply currently stands at a historical high. Hunger occurs not in regions where total food output is insufficient, but in regions where distribution systems have failed or purchasing power has collapsed. If the problem lay in "insufficient input," we should observe an absolute decline in total food output and a long-term upward trend in food prices. Neither observation holds.

Second, the global population growth rate is systematically declining. Developed countries have entered negative growth; China's total population has already passed its peak; the main contribution to global population growth now comes from regions with the lowest per capita income and the lowest per capita carbon emissions. If the predicament were attributable to "overpopulation," then the predicament should be most concentrated in regions where population growth is most rapid, and should be markedly alleviated in regions where population growth has halted. The facts are the opposite: the predicament is most concentrated and most visibly acute precisely in those regions where population growth has already stagnated or turned negative—Europe's energy crisis, East Asia's fertility anxiety, North America's polarized politics.

Third, even if we relax Malthus's original formulation and replace "food" with "resources" or "energy," the classical paradigm still misses the mark. Because it persistently locates the limit on the input side—the rate at which nature supplies matter and energy to the human system. Yet the signals of the contemporary predicament mostly issue from the output side: not "we lack sufficient energy," but "we cannot metabolize the carbon emissions produced by energy use"; not "we lack sufficient technology," but "we cannot coordinate the social consequences produced by technological application." Input-side signals (resource depletion warnings) have been repeatedly shown over the past half-century to be resolvable through price signals and synthetic substitutes; output-side signals (climate, trust, coordination) continue to accumulate and possess no price elasticity.

Locking verdict: The classical Malthusian trap mislocates the limit on the input side and is therefore unable to explain functional congestion phenomena occurring on the output side. There is a categorical mismatch between diagnosis and symptoms. Excluded.

Exclusion Two: It Is Not an Energy-Depletion-Type Crisis

The diagnosis to be excluded: The root of the contemporary predicament is the non-renewable character of fossil energy—as the era of cheap energy draws to a close, industrial civilization, built upon energy-intensive growth, will inevitably contract.

Grounds for exclusion: Explanatory redundancy, and its core prediction has been falsified.

This diagnosis gained broad academic and policy influence after the oil crises of the 1970s. Its logic is internally coherent: if a critical resource is finite and non-substitutable, then systemic growth must hit a ceiling as that resource approaches exhaustion. The problem is that its core prediction—"energy costs will rise persistently due to scarcity"—has not materialized over the past two decades.

Consider photovoltaic power. IRENA data show that between 2010 and 2020, the global levelized cost

of electricity from photovoltaics fell by approximately 85%. In a significant portion of the world, the generation cost of newly built PV plants has fallen below that of fully depreciated coal-fired plants. This is not the classical price signal of "scarcity leading to expensiveness"; it is the inverse price signal of "technological learning curves leading to cheapness."

More critically, even when energy costs were not an issue, the predicament was already there. Europe experienced an energy price shock in 2022, but even before that, during the era of cheap energy, its internal polarization, aging pressures, and governance deadlocks over immigration had been steadily accumulating. Energy is an amplifier of many symptoms, but not their generator.

This diagnosis should not be entirely discarded. The "input ceiling" phenomenon it revealed in the 1970s remains valid for specific resources (certain rare minerals, groundwater, ecosystem services). But as a total diagnosis of the contemporary global predicament, its explanatory range is entirely subsumable under Complexity Metabolic Failure—the latter can explain both the frictions of the energy transition (the shock of new energy technologies to old governance protocols) and those symptoms that continued to accumulate even during the era of cheap energy (trust decay, meta-narrative failure). Energy depletion theory is a local special case of metabolic failure, not the total etiology.

Locking verdict: Explanatory redundancy. It is logically coherent on the input side, but cannot explain the output-side symptoms that continue to intensify even when input is abundant. Excluded from the candidate set for "total diagnosis"; retained as an analytical tool for specific resource dimensions.

Exclusion Three: It Is Not Technological Stagnation Theory

The diagnosis to be excluded: The essence of the contemporary predicament is the slowdown of fundamental technological innovation—since the IT revolution, humanity has not produced another truly fundamental technological breakthrough capable of driving sustained total-factor productivity growth, and the economy has consequently entered a long stagnation.

Grounds for exclusion: Incoherence (inversion of causal direction).

The logical chain of technological stagnation theory runs: slowdown in fundamental innovation → deceleration of productivity growth → loss of economic growth momentum → intensification of social distributional conflict → manifestation of the predicament.

This chain faces refutation from two directions.

First, on the supply side, the rate of technological innovation has not slowed. Breakthroughs in AI—in natural language processing, protein structure prediction, mathematical theorem proving—the clinical translation of gene-editing technologies, the sustained advance of controlled nuclear fusion: if measured by "the quantity and quality of fundamental innovations," the past two decades do not fall short of any prior two-decade period. A critic may respond that "these innovations have not yet translated into productivity data." But this very response shifts the question: why is the translation channel from innovation to productivity blocked? This is not a problem of "no innovation," but of "the system's inability to effectively metabolize innovation."

Second, on the demand side, even if productivity data have indeed decelerated, the timing of this

deceleration does not match the timing of the outbreak of social predicaments. In the Western world, the pronounced rise of polarized politics and populist waves began after the 2008 financial crisis, whereas the so-called "long slowdown" of productivity had already been discussed by some economists two decades earlier. If technological stagnation were the ultimate cause, why was the initial phase of technological stagnation (the 1980s–2000s) recalled instead as the "golden age of globalization"? A more coherent explanation is this: technology itself has not stagnated, but the system's metabolic capacity for absorbing technology and coordinating its consequences has begun to fail. It is precisely the failure of metabolic capacity that has blocked the translation channel from innovation to productivity.

Technological stagnation theory is not entirely wrong. It correctly observes a "sense of blockage"—the system seems unable to extract the benefits of technological progress as smoothly as it once did. But it attributes the cause to the technology supply side, rather than to the system's coordination side. This is an inversion of causal direction.

Locking verdict: Incoherent. The symptom (sense of blockage) genuinely exists, but the attribution (insufficient technology supply) is mistaken. Excluded.

Exclusion Four: It Is Not Absolute Overpopulation

The diagnosis to be excluded: There are simply too many people on Earth. No matter how technology advances, the planet's carrying capacity has an absolute limit. The contemporary predicament is the result of total population approaching or even exceeding that limit.

Grounds for exclusion: Explanatory redundancy, and operationally incoherent.

This diagnosis shares the logical structure of the Malthusian trap, but shifts the focus from "food" to the vaguer notion of "planetary carrying capacity." It possesses a certain intuitive force—a finite sphere can carry only a finite population, surely.

But this intuitive force depends precisely on the vagueness of the concept. "Carrying capacity" is not a natural constant. It is a function of technology and lifestyle—the Earth's carrying capacity in the hunter-gatherer era was tens of millions; in the agricultural era, hundreds of millions; in the industrial era, billions. If we insist on an absolute ceiling independent of technology and lifestyle, we must specify the physical dimension of that ceiling. Is it space? The global population, if housed at the density of Manhattan, could fit into an area the size of a French department. Is it freshwater? Desalination technology has brought freshwater costs within agriculturally affordable range. Is it some irreplaceable mineral? Critical elements such as phosphorus and potassium do pose recycling challenges, but these challenges are not about "absolute shortage"; they are about "circular metabolism."

More importantly, at the global level, those regions with the highest per capita resource consumption and the greatest pressure on the Earth system are precisely those where population growth has already stagnated or turned negative. The per capita carbon footprint of a suburban North American resident is several tens of times that of a resident of sub-Saharan Africa. If the problem were "too many people," the solution should be to reduce the population of the regions with the highest per capita consumption—yet no serious discussion proposes that North America or Europe should "reduce its population." The discussion is always steered toward "the developing regions should control their population." This is not

a neutral scientific diagnosis; it is distributional politics draped in the mantle of carrying capacity.

From the perspective of systems ontology, "absolute overpopulation" commits a categorical error: it mistakes a regional, architectural insufficiency of metabolic capacity for an absolute surplus of total system input. The question is not how many people live on Earth, but in what manner these people organize their material and information flows. When the mode of organization (cognitive protocols, governance architecture) cannot effectively coordinate these flows, the system manifests symptoms of "overload." But the locus of these symptoms is the protocol layer, not the demographic layer.

Locking verdict: Explanatory redundancy, and relies on the non-operationalizable concept of "absolute carrying capacity." Excluded.

Exclusion Five: It Is Not a Decline in Civilizational Will

The diagnosis to be excluded: The essence of the contemporary predicament is that human civilization has lost its will to advance. We have become weak, hedonistic, short-sighted, no longer daring to take risks, no longer willing to sacrifice the present for the future. The vitality of civilization is waning.

Grounds for exclusion: Non-operationalizable, and conflicts with observable behavior.

This diagnosis belongs to a deep humanistic tradition; its lineage can be traced back to Spengler's *Decline of the West* and Toynbee's discussion of "civilizational suicide." Its emotional force lies in this: it interprets the predicament as a moral or spiritual failure, thereby summoning the individual's longing for "awakening" and "renewal."

But as a rigorous theoretical diagnosis, it confronts two fatal problems.

First, the concept of "civilizational will" cannot be measured independently of its purported "consequences." How do you know that civilizational will is declining? Because civilization is declining. Why is civilization declining? Because civilizational will is declining. This is a perfect circular argument. "Will" here is not an observable variable; it is a phantom cause retroactively projected from observed outcomes.

Second, observable behavioral data directly conflict with the narrative of "declining will." The share of global GDP devoted to R&D has risen, not fallen, over the past three decades. The total volume of entrepreneurial activity globally has never been more vigorous. Individual self-investment in education, health, and skill acquisition has reached a historical peak. Even the so-called "lying flat," when its behavioral content is carefully examined, often turns out not to be "withdrawal from all activity," but "withdrawal from activities defined by the established evaluation system, in favor of self-defined activities"—which in fact contains a powerful form of will: the refusal to be depleted by the old framework, and the persistent search for new channels of meaning.

The real problem with the civilizational will decline thesis is this: it mistakes coordination failure at the systemic level for motivational loss at the individual level. When individual effort cannot be effectively converted by the system into synergistic output, the very form of individual "striving" deforms—from effort conforming to the norms of the old protocol, to effort that deviates from or resists the old protocol. The observer sees the latter and laments the "degeneration of the times"; what the observer fails to see is

that the same life energy is still flowing—the old riverbed can simply no longer contain it.

Locking verdict: Circular argument, non-operationalizable. Excluded.

Exclusion Six: It Is Not a Single-Dimensional Crisis

The diagnosis to be excluded: The essence of the contemporary predicament is "the climate crisis" (or "the financialization crisis," "the inequality crisis," "the geopolitical crisis"—any diagnosis that assigns ultimate explanatory power to a single dimension).

Grounds for exclusion: Incomplete explanatory range; incapable of subsuming the full dimensionality of the symptoms.

If the problem were "merely" a climate crisis, then a sufficiently effective global climate agreement should suffice to alleviate the predicament. But even if we assume that all nations suddenly signed and rigorously implemented tomorrow an agreement capping warming at 1.5°C, the following symptoms would persist: endogenous volatility in financial markets, the erosion of public reason by polarized politics, the breakdown of intergenerational contracts, the governance challenges of AI, the strategic fragility of global supply chains.

If the problem were "merely" a financialization crisis, then strengthening financial regulation and curbing the disorderly expansion of capital should suffice to restore systemic health. But even if we accomplished this, carbon concentrations would continue to accumulate in the atmosphere, species loss in ecosystems would continue to accelerate, and the explanatory power of meta-narratives would continue to decay.

This is not to deny the reality or seriousness of the climate crisis or the financialization crisis. Both are real, both are serious. But establishing any one of them as the *meta-crisis*—the ultimate crisis that subsumes all others—encounters the same difficulty: in explaining crises in the other dimensions, it must rely on auxiliary hypotheses ("the climate crisis led to migration, migration led to political polarization"), and the causal chains of these auxiliary hypotheses tend to be long, reversible, and empirically riddled with counterexamples.

What is more likely to be true is this: climate, finance, inequality, geopolitics, trust, meaning—these dimensions are hitting their ceilings simultaneously. It is this very *synchronicity* that demands explanation. If they had crested sequentially, each in its own time, we could believe they merely converged on this era by chance. But their synchronous intensification points toward a deeper common dynamic—the basal process that is driving all subsystems toward their operational boundaries at the same time. That process is the thing we are seeking to name.

Locking verdict: Incomplete explanatory range. No single-dimensional diagnosis can subsume the full set of symptoms without relying on ad hoc auxiliary hypotheses. Excluded from candidacy as total diagnosis; retained as instrumental for analyzing specific dimensions.

Exclusion Seven: It Is Not the Bottom of a Cycle

The diagnosis to be excluded: The contemporary predicament is merely the normal downswing phase of a long economic cycle (Kondratiev wave) or a hegemonic cycle (Modelski long cycle). History has

always oscillated between boom and bust; this time is no different.

Grounds for exclusion: The failure of old remedies—the recovery mechanisms presupposed by cycle theory can no longer activate.

The explanatory power of cycle theory derives from the recovery mechanisms it presupposes. In an economic cycle, price signals guide resources from surplus sectors to scarce ones; in a hegemonic cycle, the rise of a challenger and the replacement of the hegemon redistribute the leadership of the international order. As long as the recovery mechanisms function, the downswing phase is merely a necessary adjustment preparing the upswing—painful, but expected to pass.

The uncanniness of the contemporary predicament lies in this: the recovery mechanisms themselves appear to be blocked.

Price signals should guide capital from fossil fuels to clean energy—they are doing so, but the speed is markedly slowed by the sunk costs of existing infrastructure, the counter-incentives of geopolitics, and social justice concerns ("who pays for the transition?"). Hegemonic transition should produce a new supplier of international public goods—but neither the rising power nor the incumbent power is willing or able to bear the full cost of providing global public goods. Monetary policy can inject liquidity, but the liquidity remains trapped within the financial system, difficult to translate into productivity growth in the real economy.

This is not the "normal pain" of a cyclical downswing phase. This is a spring that has grown fatigued—once compressed, it can no longer rebound as vigorously as before.

From a systems-theoretic perspective, cycle theory describes oscillations *within* a given architecture. The defining feature of the contemporary predicament is that the architecture itself is becoming an amplifier of oscillation rather than a damper. When the architecture itself has become the problem, the recovery mechanism is no longer a matter of "letting the cycle run its course," but of "replacing the platform on which the cycle operates."

Locking verdict: Explanatory redundancy. It is coherent under the premise that "the architecture remains functional," but cannot explain the contemporary feature of "the architecture itself failing." Excluded as total diagnosis; retained as an analytical framework for specific levels (e.g., business cycles).

Exclusion Eight: It Is Not Caused by "Bad People"

The diagnosis to be excluded: The contemporary predicament is the result of the conscious machinations or tolerance of certain specific groups—financial capitalists, globalist elites, the deep state, tech giants.

Grounds for exclusion: Excessive explanatory ambition (reliance on ad hoc hypotheses), and inability to explain the systemic distribution of the predicament.

The core logic of this diagnostic family is this: if there exists a malefactor, or class of malefactors, that can be clearly identified, then the predicament has a "prime culprit." Find the culprit, eliminate or suppress the culprit, and the predicament should dissolve.

The problem is that if the predicament were caused by "bad people," then the predicament should be most severe in those domains and regions where the power of "bad people" is strongest, and should be

markedly alleviated in those domains and regions where "bad people" have been curbed or purged. But the observational facts are these: the systemic distribution of the predicament does not significantly correlate with the geographic or sectoral distribution of any plausible "bad people" grouping. In those societies that most fiercely critique financial capital, environmental degradation and social governance difficulties are equally present. In those jurisdictions that have successfully curtailed the power of tech giants, polarized politics and meaning crises continue to deepen.

Even more fatally: suppose we replaced, overnight, every person identified as one of the "bad people" with morally unimpeachable saints. What would they discover? They would discover themselves still trapped within the same set of structural constraints—financial markets would still produce flash crashes through the coupling of algorithmic trading (even if every trader were virtuous), environmental policies would still produce unexpected compounding fallacies (even if every policymaker sincerely cared for the planet), and meta-narratives would still erode from an inability to accommodate new experience (even if every narrator honestly wished to tell the truth).

This is not to excuse misconduct. Individual greed, short-sightedness, and the will to power do exist; they do amplify the predicament and accelerate its manifestation. But they are not the etiological root of the predicament. The problem is structural: when a system's architecture makes it such that the sum of "individual rationality" necessarily converges upon "collective irrationality," moral condemnation loses its validity as a primary diagnosis.

The seductiveness of the "bad people" theory lies in its provision of a clear action script—find the enemy, defeat the enemy. To acknowledge the structural root of the predicament, by contrast, points toward a far more difficult but honest action script—we must examine and reconstruct the very architecture that forces good people into bad choices, and good policies, when superimposed, into bad outcomes.

Locking verdict: Explanatory ambition that relies excessively on ad hoc hypotheses ("they are simply too clever / too evil"), and cannot explain the persistence of the predicament when variables are controlled for. Excluded.

In summary: a system in congestion, not a will in decline. To moralize a structural problem is yet another misdiagnosis.

Table 3.1 Overview of the Eightfold Exclusion Results

No.	Excluded Diagnosis	Core Claim	Locking Criterion	What Is Retained	Feature Indicated	Constraint
1	Classical Malthusian Trap	Population growth outpaces food supply	Incoherence: aggregate food surplus; hunger is a distribution problem	As a tool of historical analysis	Constraint I (output end, not input end)	
2	Energy-Depletion Crisis	Non-renewable fossil energy causes growth to hit a ceiling	Explanatory redundancy: clean energy costs have fallen below fossil fuels; predicament accumulated even in cheap-energy era	As an analytical tool for specific resource dimensions	Constraint I (output end, not input end)	
3	Technological Stagnation Theory	Slowdown in fundamental innovation leads to long stagnation	Incoherence (causal inversion): technology supply side has not stalled; the metabolic channel for absorbing technology is blocked	Retain the observation of "blockage," correct the direction of attribution	Constraint (architectural, factorial)	II not
4	Absolute Overpopulation	Total population exceeds planetary carrying capacity	Explanatory redundancy + non-operationalizable: "carrying capacity" is a function of technology; population growth has turned negative in high-	None	Constraint (architectural, factorial)	II not

No.	Excluded Diagnosis	Core Claim	Locking Criterion	What Is Retained	Feature Indicated	Constraint
			consumption regions			
5	Decline in Civilizational Will	Humanity has lost its will to advance and its vitality	Circular argument: infers an unobservable cause from outcomes; individual striving behavior is in fact increasing	None	Constraint VI (compatibility with macro-behavior)	
6	Single-Dimensional Crisis	The predicament is "the climate crisis" or "the financialization crisis" etc. as a single dimension	Incomplete explanatory range: any single-dimensional diagnosis requires ad hoc hypotheses to subsume the full set of symptoms	Retain as tools for analyzing specific dimensions	Constraint III (explain cross-dimensional synchronicity)	
7	Bottom of a Cycle	This is the normal downswing of a long economic or hegemonic cycle	Explanatory redundancy: the recovery mechanisms presupposed by cycle theory (price signals, hegemonic transition) have failed	Retain as an analytical framework for specific levels	Constraint IV (explain the failure of old recovery mechanisms)	
8	Caused by "Bad People"	The predicament is the result of conscious machinations by specific	Excessive explanatory ambition: cannot explain the persistence of the predicament	Retain the problematic, pivot toward architectural diagnosis	Constraint V (subsume the partial truths of existing diagnoses)	

No.	Excluded Diagnosis	Core Claim	Locking Criterion	What Is Retained	Feature Indicated	Constraint
		groups	when variables are controlled for; moral attribution misses the structural root			

Table Note: The locking criteria strictly follow the dual standard of Section 3.1—incoherence (the diagnosis contains internal logical contradictions, or its premises conflict with its inferences) and explanatory redundancy (the explanatory power of the diagnosis is covered by a more parsimonious diagnosis, or it relies on non-operationalizable ad hoc hypotheses). The standard for judging whether a diagnosis is excluded is not "it is false," but "it cannot serve as a total diagnosis."

Table Function Statement

This table performs three functions within the argumentative structure:

Function One: A quick interface for tracing the argument. After the reader has finished Section 3.2, the table provides an immediate logical cross-reference. If a reader has doubts about a particular exclusion, they can precisely locate the corresponding subsection in the main text for review. The table itself is not a substitute for the argument; it is an index to the argument.

Function Two: Transparency of the locking criteria. The criterion (incoherence or explanatory redundancy) on which each exclusion is based is directly annotated in the table, making the operational standards of the locking procedure publicly visible. This precludes the misunderstanding that "the exclusions are merely the author's personal preferences."

Function Three: Logical transition to positive construction. The rightmost column, "Feature Constraint Indicated," establishes the mapping chain between the eight exclusions and the "Residual Contour" of Section 3.3. From this column, the reader can see that the six feature constraints are not proposed arbitrarily, but are the logical consequences accumulated item by item through the eight exclusions. From "what is excluded" to "what is required," the flow of the argument is transparent.

3.3 The Residual Contour After Exclusion

Table 3.1 summarizes the logical results of the eightfold exclusion. The right column lists the feature constraints pointed to by each exclusion; together, they constitute the "functional requirements" that the new concept must satisfy. We do not yet ask what it "is"—that question is left to Chapter 4—but we can ask: if a category exists that passes the exclusion procedure, what must it be able to do?

After the eightfold exclusion is complete, a negative contour emerges. We still do not know what it "is," but we do know that it must satisfy the following feature constraints:

Constraint I: It must be located on the output/metabolic end, not the input/resource end. The predicament is not "not enough of X," but "unable to metabolize the consequences produced by X."

Constraint II: It must be an architectural constraint, not a factorial one. Replacing factors (more technology, more energy, better people) cannot remove the constraint, because the constraint arises from the very mode of organization among the factors.

Constraint III: It must explain the synchronicity of symptoms—why multiple dimensions such as climate, finance, trust, and meaning are simultaneously approaching their operational boundaries. Synchronicity implies that what is being approached is not multiple independent limits, but a single basal limit differentially manifested across different subsystems.

Constraint IV: It must explain the failure of old recovery mechanisms—why the rebound predicted by cycle theory has not appeared, why the efficacy of price signals, hegemonic transitions, and policy interventions is attenuating.

Constraint V: It must be able to accommodate the partial truths captured by existing single-dimensional diagnoses (involution, limits to growth, systemic risk, governance crisis), while simultaneously subsuming the residual symptoms that each of them is unable to explain independently.

Constraint VI: It must be compatible with observable macro-behavior: technology continues to advance, population growth is decelerating, individual will is seeking new outlets—yet the coordination efficacy of the system as a whole is decaying.

These six feature constraints constitute the "functional requirements" for the new concept. Any category claiming to explain the contemporary global predicament must satisfy all six simultaneously; if it cannot, it is a candidate that has been negated by the exclusion procedure of this chapter.

Table of Exclusion-to-Constraint Mapping

Exclusion Item	Where It Breaks Down	Feature Constraint
Classical Malthusian	Mislocates limit on input end	Constraint I (locate on output/metabolic end)
Energy Depletion	Explanatory redundancy	Constraint I

Exclusion Item	Where It Breaks Down	Feature Constraint
Theory		
Technological Stagnation	Causal inversion	Constraint II (architectural, not factorial)
Absolute Overpopulation	Explanatory redundancy + non-operationalizable	Constraint II
Civilizational Will Decline	Circular argument	Constraint VI (compatibility with macro-behavior)
Single-Dimensional Crisis	Incomplete explanatory range	Constraint III (explain cross-dimensional synchronicity)
Bottom of a Cycle	Explanatory redundancy	Constraint IV (explain failure of old recovery mechanisms)
"Bad People"	Excessive explanatory ambition	Constraint V (subsume partial truths of existing diagnoses)

3.4 Self-Checklist of Residual Possibilities

In accordance with the openness commitment of the Abductive Locking Method, if a reader believes that a ninth type of position exists that has not been excluded, it should at minimum be able to clearly answer the following questions:

- Can the position be stated as a clear core proposition (not exceeding one paragraph)?
- Is the "constraint location" identified by the position on the input end or the output end?
- Can the position explain the cross-dimensional synchronicity of symptoms (the simultaneous hitting of ceilings by climate, finance, trust, and meaning)?
- Can the position explain the collective attenuation of old recovery mechanisms (price signals, cyclical rebounds, policy interventions)?
- Can the core variable of the position be operationalized—that is, do there exist, at least in principle, independent observational indicators for it?
- Does the explanatory power of the position depend on non-operationalizable ad hoc hypotheses (such as "the extraordinary evil of a certain group" or "a mysterious historical destiny")?

If there exists an alternative position that passes all six of these questions, the locking procedure of Chapter 3 must be reopened. This is precisely the self-corrective mechanism of the Abductive Locking Method.

Chapter Summary

Chapter 3 has executed the core stage of the Abductive Locking procedure—demarcating the boundary of "what it is not" through an eightfold exclusion.

We have examined, one by one, eight possible diagnoses that occupy prominent positions in contemporary public discourse and academic research: the classical Malthusian trap, the energy-depletion crisis, technological stagnation theory, absolute overpopulation theory, civilizational will decline theory, single-dimensional crisis theory, the bottom-of-a-cycle theory, and the conspiracy or "bad people" theory. Each exclusion has been conducted on the grounds of "incoherence" or "explanatory redundancy."

After the eightfold exclusion is complete, a negative contour has emerged: the thing we are looking for must be located on the output/metabolic end; must be an architectural rather than a factorial constraint; must be able to explain the cross-dimensional synchronicity of symptoms; must be able to explain the failure of old recovery mechanisms; must be able to subsume the partial truths of existing single-dimensional diagnoses; and must be compatible with observable macro-behavior.

These six feature constraints constitute the "functional requirements" for the concept construction of Chapter 4. We now reverse direction into positive construction—what is the form of the category that satisfies all these constraints? What is its formalized definition? What is its kernel postulate, irreducible any further?

Chapter 4: The Conceptual Kernel: The Formal Construction of Complexity Metabolic Failure

Chapter 3 completed the exclusion procedure. After the eightfold negation, a residual contour emerged: the thing we are looking for must be located on the output end, not the input end; must be an architectural constraint, not a factorial one; must explain the cross-dimensional synchronicity of symptoms; must explain the failure of old recovery mechanisms; must subsume the partial truths of existing single-dimensional diagnoses; must be compatible with observable macro-behavior.

The task of this chapter is the positive construction of the category that satisfies all six constraints. We proceed sequentially: formalized definition (4.1), kernel postulate (4.2), isomorphic model (4.3), and a necessity test under Occam's Razor (4.4).

4.1 Formalized Definition

Based on the exclusion constraints of Chapter 3 and the conceptual preliminaries of Section 1.4, we now provide the formal definition of Complexity Metabolic Failure.

Complexity Metabolic Failure (CMF):

The state of functional congestion—characterized by valid input but disordered output—into which a noospheric system enters when the total complexity (C) generated by the horizontal interactions among its subsystems through self-organizing evolution persistently exceeds, over a given observation period, the information-processing and coordination bandwidth ceiling (P) of the system's top-level governance protocol. The irreversibility of this state derives from a structural asymmetry: subsystem interaction complexity is a combinatorial function of the number of subsystems, while the growth of protocol metabolic capacity is subject to the physical and logical limits of information processing.

This definition contains the following operationalized components:

C: Interactive Complexity Output. The total incremental rate of novel interactive relations, unintended consequences, and coordination demands generated among subsystems within a system as a result of self-organizing evolution. Its growth is not a linear function of the number of subsystems, but a combinatorial function of the number of potential interactions among them— n subsystems can generate at most $n(n-1)/2$ bilateral interactions, and each interaction can generate unpredictable consequences across multiple dimensions. *C* measures the information increment of these interactions, not the number or scale of the subsystems themselves.

P: Protocol Metabolic Capacity. The upper limit of information-processing capacity by which the system's top-level governance architecture can, per unit time, effectively interpret, coordinate, and respond to subsystem interaction events. It comprises bandwidth constraints across three links: perception (whether an interaction event can be effectively registered and brought into the center of attention); interpretation (whether the causal structure of the interaction event can be correctly attributed and understood); and response (whether effective coordinating actions can be generated and executed). A bandwidth insufficiency in any one of these three links constitutes a bottleneck on *P*. The total throughput of the three determines the effective value of *P*, with the minimum of the three serving as the hard upper limit of the system's protocol metabolic capacity.

Functional Congestion: When the gap $C > P$ persists, the system does not immediately cease to operate—subsystems continue to function, resources continue to flow in, output continues to be produced. But for each additional unit of input (new rules, new technologies, new information), the net gain in total system efficacy is no longer positive, but zero or negative. The system enters a dynamic phase distinct from collapse: it is still running, but it can no longer advance.

4.2 Kernel Postulate

The formalized definition is operationalizable, but it contains multiple concepts that require further argumentation. We reduce these concepts further, until only a single core proposition remains that cannot be further reduced.

Kernel Postulate: For any system constituted by subsystems possessing self-organizing capacity, there exists a necessary time lag and bandwidth differential between the information-processing capacity of its top-level governance protocol and the information-generation rate of the horizontal interactions among its subsystems. When the information-generation rate persistently exceeds the information-processing rate, functional congestion is irreversible.

The power of this postulate lies in this: to refute it requires denying certain more fundamental conditions of existence.

To deny the "time lag" requires denying causality. Information processing involves physical processes: perception requires signal input, interpretation requires computation time, response requires action execution. To deny the existence of time lag is equivalent to claiming that causal relations can be fully interpreted in zero time—which is impossible in any finite system.

To deny the "bandwidth differential" requires denying combinatorial explosion. The number of potential subsystem interactions grows combinatorially with the number of subsystems. For information-processing capacity to always cover the information-generation rate, the system would need, at every moment, unlimited perceptual, interpretive, and response capacities—which is equivalent to assuming an infinite material and energetic substrate.

To deny "irreversibility" requires denying the Second Law of Thermodynamics. Once an interaction has occurred, it has left causal traces in the system—newly formed dependency structures, newly entrenched vested interests, newly deposited cognitive habits. These traces cannot be "erased" as though they had never occurred. The amount of information processing required to clear a metabolic deficit is always greater than the amount that would have been required to process it promptly from the start—this is the irreversible entropy increase of metabolism.

The kernel postulate is therefore not an empirical hypothesis susceptible to easy falsification by empirical facts, but a structural truth approaching tautology: if the physical limits of information processing hold, if the combinatorial growth of subsystems holds, if the ineradicability of causal traces holds—then $C > P \rightarrow$ functional congestion follows as a necessity.

4.3 Isomorphic Model: The Coupling Failure of Windows XP and Its Applications

To help the reader grasp the relation between C and P in a concrete scenario, we introduce an isomorphic model drawn from computer systems.

4.3.1 Basic Mapping of the Model

Consider human civilization as a running computer:

The parent system: the computer's operating system. It defines the rules for communication among applications, the protocols of resource allocation, and the mechanisms for error handling. In civilization, this is the top-level governance protocol—including laws, institutions, international treaties, and, at a deeper level, the meta-narrative.

The subsystems: the applications running on the operating system. Each application independently optimizes its own functionality. In civilization, these are the financial system, technological R&D, the energy industry, information platforms, trade networks, and so forth.

Interactive Complexity (C): the horizontal communication traffic generated among applications through data exchange, API calls, file sharing, and the like. This communication is not pre-planned by the operating system—it emerges spontaneously as the applications run. In civilization, this is the cross-domain interactions: finance investing in technology, technology restructuring employment, energy rewriting geopolitics, information eroding consensus.

Protocol Metabolic Capacity (P): the throughput bandwidth of the operating system's I/O interface, along with the real-time monitoring and coordination capacity of the task manager. In civilization, this is the sum of cross-departmental coordination mechanisms, international multilateral platforms, and spaces for public deliberation.

4.3.2 The Evolution of Conflict: A Temporal Sequence

Phase One: Architectural Compatibility. When the operating system (XP) was first released, the applications were mainly document processors, email clients, and basic tools. Horizontal communication traffic among applications was minimal—a document processor and an email client did not engage in frequent data exchange. The I/O bandwidth of the operating system far exceeded the communication traffic. The system ran smoothly.

Phase Two: Subsystem Upgrading. Applications undergo independent iteration. Video editing software, database management systems, and real-time communication tools are installed. Each application becomes more powerful—processing larger files, executing more complex computations, producing richer outputs. The "vertical capability" of each individual application improves markedly.

Phase Three: The Emergence of Horizontal Coupling. Novel patterns of horizontal communication begin to emerge. The output of video editing software is imported into presentation software; the presentation is sent to a real-time communication group; the discussion in the group is entered into a database; database queries trigger automatic email replies. These interactions were not planned by the operating system's designers—they are spontaneous couplings among applications. The I/O bandwidth of the operating system begins to feel the pressure—occasional stutters and lags appear.

Phase Four: Metabolic Failure. Applications continue to upgrade. A newly installed AI application requires high-frequency, high-volume bidirectional communication with nearly all the old applications. The horizontal communication traffic exceeds the I/O bandwidth ceiling of the operating system. What now appears is an uncanny systemic behavior: the hardware resources (CPU, memory, hard disk) are not exhausted—the system monitor shows that physical resources still have redundancy—but the operating system can no longer effectively coordinate the interactions of all the applications. Applications begin to "go their own way"—bypassing the standard interfaces of the operating system, establishing unofficial direct communication paths. The overall behavior of the system becomes increasingly unpredictable. Collapse has not yet occurred, but functional congestion is already in full swing.

Phase Five: The Terminal Stage at the Meta-Narrative Level. At the civilizational level, the final signal of functional congestion is this: the designers of the operating system (the bearers of the meta-narrative) no longer believe that the system can be repaired. They stop trying to coordinate the conflicts among applications and instead each chooses to "abandon the operation of some applications"—withdrawing from traditional political participation, from global cooperation agreements, from intergenerational ethical responsibility. The operating system appears, on the surface, still to be running—but its legitimacy narrative has decayed. The system has entered a state of uncoordinated inertial drift. This is not collapse—collapse is the cessation of operation. Drift is the continuation of operation, but without any coordination center still capable of comprehending the whole and responding consciously.

4.3.3 Validity Statement and Limitations of the Model

This isomorphic model is not an argument; it is a ladder for cognition. Its validity lies in this: it helps the reader "see" the dynamic relation between *C* and *P* within a familiar, concrete scenario. Its limitations are equally clear:

- A civilizational system is more complex than a computer system—subsystems possess not only self-organizing capacity but also survival will and strategic behavior. They do not automatically pause their upgrading when they encounter coordination bottlenecks.
- The upgrading of an operating system is typically accomplished by a single agent (Microsoft), whereas the "operating system upgrade" of civilization cannot be accomplished by any single agent—this is precisely the difficulty of the phase transition.

Having passed through this model, the reader may return to the formalized definition of Section 4.1 and independently test its logical quality at the level of abstract concepts.

4.4 The Necessity Argument: The Occam's Razor Test

For a concept to earn its right to exist in the academic world, it must pass the test of Occam's Razor: is it more parsimonious and more necessary than the existing combination of categories?

As diagnosed in Section 1.2, there are four main categories deployed in contemporary public discourse for explaining the global predicament: *involution*, *limits to growth*, *systemic risk*, and *governance crisis*. If their conjunction were capable of fully subsuming all the symptoms exhibited in Section 1.1 of this paper, then Complexity Metabolic Failure would be a redundant coinage—and the Razor should cut it down.

4.4.1 The Coverage Blind Spots of the Existing Categories

Involution: Explains diminishing marginal returns within a single dimension, but cannot explain the compounding fallacies generated by cross-dimensional interactions. In the contemporary predicament, many of the most severe damages occur at the boundaries between dimensions—environmental protection and food, finance and the real economy, information and consensus—not within any single dimension.

Limits to Growth: Explains physical ceilings on the resource inflow end, but cannot explain metabolic congestion under conditions of resource abundance. Carbon emissions are not a problem of "not enough fossil energy," but of "the climate system cannot metabolize carbon emissions." The predicament is on the output end.

Systemic Risk: Explains cascading contagion with an identifiable triggering event, but cannot explain sustained smoldering without a triggering event. Since 2008, the global system has not experienced another "Lehman Moment" of comparable magnitude, yet financial fragility, social polarization, and the decay of international trust have continued to deepen.

Governance Crisis: Attributes the predicament to the failure of institutions or decision-makers, but cannot explain why the best-intentioned policies, when superimposed, produce synthetic fallacies; nor can it explain why even highly transparent governance systems stumble just as much.

4.4.2 The Razor Test: The Logic and Economy of One Replacing Four

Now ask: to cover the full set of symptoms described above, what is the minimum number of independent concepts required?

If we use the conjunction of existing categories: we need to deploy simultaneously the four categories of "involution + limits to growth + systemic risk + governance crisis," and to establish among them a set of ad hoc coordination rules—for example, "governance crisis amplifies systemic risk, systemic risk compounds limits to growth, limits to growth induce involution." This coordination rule itself requires independent argumentation, and its causal chain is empirically riddled with counterexamples.

If we use Complexity Metabolic Failure: a single category. Its differential manifestations across four dimensions—involution as its single-dimensional projection, the limits-to-growth input signal as its misreading, systemic risk as its event-release form, governance failure as its architectural root—are all logical consequences derivable from the same dynamic inequality ($C > P$), requiring no ad hoc auxiliary hypotheses.

This is not to say that involution, limits to growth, systemic risk, and governance crisis are "false"—each captures a real facet of the symptoms. But the residual symptoms that each independently fails to explain are precisely what constitute the explanatory object-domain of Complexity Metabolic Failure. To replace the independent superposition of four concepts with a single concept is, theoretically, the more parsimonious and more economical solution.

Table 4.1 Occam's Razor Test: Single Concept vs. Four-Category Conjunction

Phenomenon Requiring Explanation	Involution	Limits to Growth	Systemic Risk	Governance Crisis	Complexity Metabolic Failure
Diminishing marginal returns within a single dimension	✓	—	—	—	✓ (as a special case)
Cross- dimensional compounding fallacies	X	—	—	—	✓
Disordering under material abundance	—	X	—	—	✓
Metabolic congestion on the output end	—	X	—	—	✓
Cascading collapse with a	—	—	✓	—	✓ (as a special case)

Phenomenon Requiring Explanation	Involution	Limits to Growth	Systemic Risk	Governance Crisis	Complexity Metabolic Failure
triggering event					
Sustained smoldering without a triggering event	—	—	X	—	✓
Compounding fallacies of well- intentioned policies	—	—	—	X	✓
Collective failure of old recovery mechanisms	—	X	—	X	✓

Table Note: ✓ indicates that the category can independently (without reliance on ad hoc auxiliary hypotheses) explain the phenomenon; — indicates that the category does not claim to explain the phenomenon; X indicates that the category claims to explain the phenomenon but suffers from logical incoherence or explanatory leakage. The argumentative function of this table is not to "prove that CMF is correct," but to demonstrate that, without CMF, one must deploy at least four independent concepts in parallel, supplemented with a set of ad hoc coordination hypotheses, to achieve the same subsumption—whereas CMF achieves this subsumption with a single postulate. This is a textbook application scenario for Occam's Razor.

4.4.3 Razor Verdict

Complexity Metabolic Failure passes the Occam's Razor test. It is not a supplementary perspective alongside the existing categories, but their subsumptive postulate. It is theoretically necessary—not because it is "more correct" than the existing categories, but because it explains, with a simpler postulate, the same set of phenomena that the existing categories can each explain only partially.

The work of concept construction is hereby complete. Chapter 4 has provided the formalized definition, the kernel postulate, the isomorphic model, and the necessity argument. On this foundation, Chapter 5 will proactively disclose the boundaries and failure conditions of this concept—for a concept that cannot say where it fails has not yet completed its final academic legitimization.

Chapter 5: Boundaries and Failure Criteria: The Self-Defense of the Concept

A concept that declares only "under what conditions it holds" but never declares "under what conditions

it fails" is logically equivalent to one that "holds under all conditions"—and a proposition that holds under all conditions is not a scientific concept; it is a dogma.

The task of this chapter is to proactively disclose three aspects of the boundary of Complexity Metabolic Failure:

- (1) Its ontological boundary—under what philosophical limit assumptions the dynamic state described by this concept would dissolve or deform;
- (2) Its operational boundary—under what observable conditions we should judge that "metabolic failure is occurring" rather than "some other problem is occurring";
- (3) Its draft criteria—laying the tracks from "qualitative diagnosis" to "measurable indicators" for subsequent quantitative testing.

This is not a theoretical self-weakening. Quite the contrary: only when a concept clearly states "where it ceases to hold" does it acquire unshakeable stability within the domain where it still does hold.

5.1 Ontological Boundary: Where the Concept Fails

What we ask is not "what factors might alleviate metabolic failure"—that belongs to the domain of "solutions," discussed in Chapter 7. What we ask is a more fundamental question: under what conditions would the necessary relation itself—"Interactive Complexity Output > Protocol Metabolic Capacity → Functional Congestion"—dissolve?

The answer is: at least one of the following two conditions must be satisfied.

5.1.1 Condition One: Interactive Complexity Output Is Reduced to Zero

If the horizontal interactions among subsystems no longer generate any new information—that is, whenever the output of subsystem A couples with the input of subsystem B, the mode of coupling is always already fully encoded and anticipated by the existing protocols, with no further unintended consequences, emergent properties, or interaction patterns that have yet to be assigned a place—then, no matter how low the protocol metabolic capacity, the inequality $C > P$ can never hold, because $C = 0$.

This amounts to requiring a social system to enter a state of absolute interactive stasis: new technologies no longer alter relations of production, new information no longer alters public opinion, new trade no longer alters geopolitical configurations, new ideas no longer challenge old narratives. Subsystems continue to operate, but every interaction among them is an exact repetition of the past. Nothing new is born from interaction.

This condition is logically conceivable, but physically equivalent to denying the self-organizing property of complex systems. The self-organization tendency—the intrinsic propensity of system elements, in the course of their interactions, to spontaneously generate more complex structures—is the fundamental dynamic of existence stated in Meta-Axiom Zero of *Systematic Ontology*. To make $C = 0$, one would have to completely suppress all self-organizing processes within the system through external coercion—yet the suppression itself would consume information-processing capacity, thereby *increasing* C instead. This is a self-defeating condition: to eliminate complexity, you must impose regulatory complexity; the regulatory complexity itself becomes a new source of interaction complexity.

Conclusion: Condition One is not instantiable in human civilizational systems. It does, however, provide the first clear boundary of the concept—if, in the future, there were to exist some hypothetical system-form in which all interactions were fully deterministic and zero-unintended-consequence, then the concept of Complexity Metabolic Failure would not apply to that system. This is not a failure of the concept; it is a declaration of the concept's domain of applicability.

5.1.2 Condition Two: Protocol Metabolic Capacity Tends Toward Infinity

If there existed a coordinator—we borrow the metaphor of Laplace's demon—capable of perceiving, in zero time delay, the full information of all subsystem interactions; of completing, in zero time delay, the interpretation of all the causal consequences of each interaction event; and of generating and executing, in zero time delay, the optimal coordinating response—then P would tend toward infinity. Under such conditions, no matter how large C , the inequality $C > P$ could never hold. Metabolic failure could not occur.

This condition is logically conceivable. But physically, information processing requires a material substrate, and the state changes of a material substrate are subject to fundamental physical constants—the speed of light sets the minimum time delay for information transmission; the Planck constant sets the finest granularity for information encoding; Landauer's principle sets the minimum energy cost for information erasure. Infinite information-processing capacity requires an infinite material substrate—infinite mass, infinite energy, or infinite time. Any of these three "infinities" contradicts the physical fact of a finite universe.

Moreover, even if we set aside physical limits and proceed from systems theory alone, the very assumption of "infinite protocol metabolic capacity" dissolves the premise of the concept of "system" itself. A system is a system precisely insofar as it has boundaries—material boundaries, energetic boundaries, information-processing boundaries. A "system" without boundaries is conceptually equivalent to the universe itself. And the universe itself, according to the statement of Meta-Axiom One of *Systematic Ontology*, can maintain its stable existence precisely because all subsystems within it have boundary constraints. A "Leviathan" possessing infinite coordinating power is either a logical God or a thermodynamic impossibility.

Conclusion: Condition Two is physically non-instantiable. It does, however, provide the second clear boundary of the concept—if, in the future, there were to exist some infinite coordinator (for example, an omniscient and omnipotent AI-god), the concept of Metabolic Failure would not apply to it. The practical significance of this boundary is this: any technological solution or institutional design that claims to be able to "solve metabolic failure once and for all" logically presupposes an impossible "infinite coordinator." Recognizing this presupposition is itself a diagnostic function of the concept of Metabolic Failure.

5.1.3 The Theoretical Significance of the Boundary Declaration

The concept fails only under the limit conditions of "interactive complexity reduced to zero" or "coordinating power tending toward infinity." Both limit conditions are logically conceivable but physically non-instantiable. This means: across all possible evolutionary paths of human civilization, the

dynamic constraint of Complexity Metabolic Failure is structural—it is not a contingent historical phase, but an inevitable constraint that any complex noospheric system possessing self-organizing subsystems must confront.

This is not pessimism. To recognize that a constraint is structural means precisely this: you cannot solve the problem by waiting for it to go away on its own. You must, while acknowledging the constraint, search for methods to improve metabolic efficiency within it. This is like the Second Law of Thermodynamics not being "pessimistic," but being a boundary condition—it does not tell you how high you can fly, but it tells you that a perpetual motion machine is impossible, thereby directing the researcher's attention toward the genuinely operationalizable efficiency zone.

5.2 Draft Operational Criteria

With the ontological boundary of the concept established, the next question is: back in the real world, how do we know whether a system is entering metabolic failure? If a concept cannot, even in principle, be observed, it remains suspended in an abstract cloud.

This section proposes three draft criteria. They need not, at this stage, possess full quantifiability—that is the task of subsequent empirical research—but they must be sufficiently clear to tell us: "If this concept holds, what should we observe?"

5.2.1 Criterion A: The Growth Rate of Macro-Coordination Costs Exceeds the Growth Rate of Utility (the Reversal-of>Returns Criterion)

Statement of the Criterion: Over a given observation period, the total costs expended by the system to maintain overall coordination functions (including but not limited to: the costs of rule-making and rule-revision, the costs of compliance and supervision, the costs of conflict mediation and dispute resolution, the total duration and total person-hours of cross-departmental coordination meetings) grow persistently faster than the total utility of the system.

Observable Signals:

- The revision frequency and page length of regulations and policies show an upward trend, but the "marginal contribution to social satisfaction" of each newly added regulation shows a downward trend;
- The share of full-time coordination roles (administrators, compliance officers, mediation bodies) in total labor time rises persistently, yet organizational efficacy indicators (such as the delay from decision to execution) show no improvement or even deteriorate;
- The frequency of "crisis-response meetings" rises, but the conversion rate of meetings into effective actions declines, with meeting output increasingly consisting of "arranging the next meeting."

Theoretical Basis: When protocol metabolic capacity is surpassed by interaction complexity, the system is compelled to invest ever more resources into coordination activities themselves, merely to maintain the appearance of coordination. These investments may initially maintain a stable level of efficacy (the saturation phase), but as the *C/P* ratio continues to rise, the marginal utility of incremental investment turns negative—this is precisely the signature signal of functional congestion at the observable level.

5.2.2 Criterion B: The Response Cycle's Time Lag Exceeds the Crisis Half-Life (the Time-Lag Deadlock Criterion)

Statement of the Criterion: Over a given observation period, the typical response time lag of the system to key interaction events (from the event being first perceived to the execution of coordinating action) persistently exceeds the "crisis half-life" of that class of events—that is, the time it takes for the negative consequences of the event, if left untreated, to spontaneously diffuse to the entire system.

Observable Signals:

- The review cycle of financial regulatory agencies for novel trading strategies > the cycle within which such strategies are replicated at scale in the market and generate systemic risk;
- The regulatory cycle of environmental legislation for novel pollutants > the cycle within which such pollutants diffuse through ecosystems to the irreversible threshold;
- The response cycle of information ecology governance to novel forms of disinformation > the cycle within which such information forms spread from the fringe to become a widely held social consensus;
- The response cycle of international treaties to novel forms of geopolitical conflict > the cycle within which such conflicts escalate from localized frictions to full-scale confrontation.

Theoretical Basis: Time lag is the direct proxy variable of protocol metabolic capacity. When the total delay across the three links of perception, interpretation, and response persistently exceeds the diffusion speed of the event, the system is causally incapable of preventing the accumulation of negative consequences. The deficits from each round of responses lagging behind their corresponding diffusions do not disappear—they enter the next round as unprocessed backlogs. This is the temporal manifestation of $C > P$.

5.2.3 Criterion C: The Explanatory Coverage Rate of the Meta-Narrative Undergoes Irreversible Decline (the Antagonistic Failure Criterion)

Statement of the Criterion: The explanatory coverage rate of the system's top-level meta-narrative—the fundamental collective narrative that explains "who we are, why we are here, where we are headed"—shows a monotonic decreasing trend over a cross-generational observation period, and no alternative narrative within the system achieves a comparable level of consensus coverage.

Observable Signals:

- The net sentiment score of core narrative elements (such as "progress," "freedom," "growth," "rejuvenation") in public discourse declines persistently, and this decline does not recover with the fluctuations of economic or electoral cycles;
- The number of mainstream explanatory frameworks for the same event increases—consensus differentiates at the level of "variety"—but no single framework can independently secure majority endorsement;
- The rate of identification with established core narratives among younger cohorts is systematically lower than that of the preceding generation at the same age (intergenerational decay);
- "Exit" behaviors—withdrawal from traditional political participation, traditional consumption patterns, traditional career trajectories—show a statistically rising trend across classes and regions, and those who exit do not provide clear new narratives about "where they are exiting

to."

Theoretical Basis: The meta-narrative is the "cognitive skeleton" of the system's top-level governance protocol. It not only coordinates behavior but also provides the meaning-legitimacy of behavior. When the genuinely novel experience produced by subsystem interactions (new technologies, new social relations, new ecological facts) persistently exceeds the range that the old narrative can explain and assign a place to, the narrative must continuously engage in self-justification—and each act of justification consumes some of the narrative's credibility. When this consumption exceeds the narrative's own repair rate (the resources and affect invested by the narrative community in maintaining the narrative), the explanatory coverage rate of the narrative enters irreversible decline. This decline is the terminal signal of metabolic failure as it penetrates from the "functional level" into the "meaning level."

The Dynamic Explanation of Antagonistic Failure: According to the "Rights Anchor Theorem" and its corollary in *Noospheric Ecology*, a healthy system relies on the sustained antagonism exercised by its components (individuals, communities, media, academia) against subsystem alienation to maintain protocol effectiveness. When the explanatory power of the meta-narrative decays below a threshold, the antagonistic forces themselves lose their narrative anchor—individuals no longer believe that "if I protest/criticize/correct, the world will become better"—and thus cease to invest in antagonistic behavior. Antagonistic failure is the collapse of protocol metabolic capacity on the meaning dimension: it is no longer merely slow; it is no longer believed to be worth doing.

5.2.4 The Relations Among the Three Criteria

The three criteria are not three independent "tests," but the differential manifestations of metabolic failure at three levels:

- Criterion A (Reversal of Returns) tests the system's *efficiency layer*—whether the input-output ratio has turned from positive to negative.
- Criterion B (Time-Lag Deadlock) tests the system's *time layer*—whether the system's response clock has become slower than the crisis's diffusion clock.
- Criterion C (Antagonistic Failure) tests the system's *meaning layer*—whether the system's self-corrective mechanisms have lost their operational drive because their narrative foundation has collapsed.

Causal relations obtain among the three levels: the deterioration of the efficiency layer (Criterion A) makes the problem of response lag difficult to resolve through increased investment (Criterion B worsens); the persisting deadlock at the time layer continuously wears away narrative credibility; the collapse of the meaning layer (Criterion C) eventually renders any improvement of the efficiency layer impossible—because no one any longer believes that "improvement" itself is possible.

This is not a mechanical temporal sequence. The deterioration of the three levels can overlap; they can accelerate one another. But when all three criteria simultaneously test positive, we may judge with sufficient confidence that the system has entered structural metabolic failure—not a transient manifestation of a cyclical downturn, not a local fault in some particular subsystem, but a state in which the architecture itself has become the bottleneck.

Summary Table of Relations

Criterion	Layer Tested	Core Signal	Deterioration Path
A (Reversal of Returns)	Efficiency layer	The marginal return of coordination investment turns negative	→ Increasing investment can no longer solve the problem
B (Time-Lag Deadlock)	Time layer	Response is slower than crisis diffusion	→ Accumulation of unprocessed metabolic deficits
C (Antagonistic Failure)	Meaning layer	The narrative foundation of the will to correct collapses	→ Any improvement of Criterion A is no longer possible

5.3 Operationalization Roadmap for the Criteria

This section lays the tracks for future empirical researchers from "draft criteria" to "operationalizable measurement." What follows is not a complete measurement protocol, but an indication, for each criterion, of the direction of operationalization and the currently available proxy data sources.

Direction of Operationalization for Criterion A

"Macro-coordination costs" as a direct metric does not exist in the current national accounts systems of statistical agencies. However, the following proxy indicators can be established:

- The annual growth rate of total word count of legal texts (representing the volume of rule production) compared to the growth rate of per capita GDP;
- The trend in the share of managerial and administrative occupations in total employment;
- The trend in the share of corporate compliance costs in total operating costs;
- The long-term trajectory of "economic policy uncertainty indices" (indicators already supported by a mature academic literature).

When the systematic upward trends of the above indicators persist beyond a full economic cycle (typically 7–10 years) and cannot be fully explained by cyclical factors, Criterion A may be regarded as positive.

Direction of Operationalization for Criterion B

Event-response time lags require subsystem-specific, targeted measurement. Operationalizable paths include:

- Selecting a number of historically completed typical "response failure" cases (such as the legislative delay for a particular environmental pollutant, the regulatory delay for a particular financial instrument), calculating the duration from "first expert warning" to "formal regulation taking effect," and observing whether this duration exhibits a systematic lengthening trend across domains;
- At the level of international treaties, calculating the average cycle from "initiative proposed" to "entry into force" for multilateral agreements, and observing its trend.

A positive determination for Criterion B requires that the lengthening trend of response cycles cannot be

fully explained by the single factor that "the problems themselves are becoming more complex"—for "problems becoming more complex" is itself a manifestation of interactive complexity output, and belongs to the *explanandum*.

Direction of Operationalization for Criterion C

The operationalization of the explanatory coverage rate of the meta-narrative can draw on existing social surveys and text-analysis tools:

- Using sentiment analysis tools assisted by large language models, conduct long-term tracking of the net sentiment scores of core narrative symbols in major media texts;
- Using generational panel data, compare the identification rates with core narratives of different cohorts at the same life stage;
- Track the long-term trends of "exit behaviors" in the labor market (the share of non-traditional employment forms), in social participation (voting rates, association membership rates), and in consumption patterns.
- Utilize the cognitive escape difficulty index derived from Theorem D23 of *Noospheric Ecology*:

$$D_c = \gamma \cdot \frac{A^2 \cdot N_i}{M_a}$$

where A is the degree of paradigm authority (proxied by academic citation concentration or media discourse power), N_i is the proportion of the population that has internalized the paradigm (obtainable through social surveys), and M_a is the average metacognitive capacity of the society (estimable through educational levels and critical thinking test scores). When D_c rises persistently and is accompanied by a fragmentation of explanatory frameworks, Criterion C is positive.

Preliminary Empirical Corroboration

Pending formal large-scale empirical testing, a number of macro-signals consistent with the direction of the three criteria can already be observed:

- The World Economic Forum's *Global Risks Report* has for multiple consecutive years ranked "social polarization" and "multilateralism in decline" as high-probability risks, and the rankings of these risks in the report have continued to rise—this is consistent with the expectations of Criterion B (Time-Lag Deadlock) and Criterion C (Antagonistic Failure).
- Multiple cross-national social surveys indicate that the expectation among younger generations that "their children will live better than their parents" has fallen to a historical low in multiple developed countries—this is a direct signal of Criterion C (intergenerational decay of meta-narrative explanatory power).
- The rapid growth of the global RegTech market itself contains a reflexive signal: the system needs ever more technological tools to cope with the complexity of compliance and supervision, and this is an indirect manifestation of Criterion A (the growth rate of coordination costs exceeding that of utility).

These are not "proof," but "directional corroboration." They constitute the empirical anchors between the concept construction of this paper and subsequent empirical testing.

Chapter Summary

Chapter 5 has completed the final step of the concept construction procedure—the proactive disclosure

of the concept's boundaries and criteria.

Section 5.1 specified the ontological boundary of the concept: Complexity Metabolic Failure dissolves only under the two philosophical limit assumptions of "interactive complexity reduced to zero" or "protocol metabolic capacity tending toward infinity." Neither of these assumptions is instantiable in human civilizational systems, thereby establishing the structural stability of the concept within the phase space of human evolution.

Section 5.2 proposed three draft operational criteria: Reversal of Returns (Criterion A), Time-Lag Deadlock (Criterion B), and Antagonistic Failure (Criterion C). The three criteria correspond respectively to the efficiency layer, the time layer, and the meaning layer of the system, constituting the tracks from "qualitative concept" to "observable signals."

Section 5.3 provided the operationalization direction and currently available empirical indicators for the criteria, laying down clear pathways for subsequent empirical research.

The concept has been constructed, its boundaries have been demarcated, and its criteria have been specified. Now we must return to the map of academic knowledge and answer the question of how this new concept relates to existing paradigms. This is the task of Chapter 6.

Chapter 6: Dialogue and Delimitation: The Theoretical Ecological Niche of the Concept

That a concept is logically coherent, procedurally legitimate, and clear about its boundaries does not yet mean its academic mission is complete. It must also answer an ecological question: within the existing map of human knowledge, what territory do you occupy? To which theoretical family do you belong, and from which superficially similar relatives must you be rigorously distinguished?

This chapter performs that task. We do not repeat the exclusion arguments of Chapter 3—the logic of Chapter 3 was "why the old categories cannot serve as total diagnoses." The logic of this chapter is "now that the new concept has been erected, what are its relations of inheritance, revision, or replacement with respect to the old categories?" These are two different kinds of dialogue: the former is passive defense (proving one is not superfluous); the latter is active demarcation (precisely specifying one's theoretical ecological niche).

The objects of dialogue are, in order: the Malthusian trap, the Limits to Growth thesis, the concept of involution, and systemic risk theory. These four constitute the theoretical lineage most directly relevant to Complexity Metabolic Failure. A summary table at the chapter's end will consolidate all the results of this demarcation.

6.1 With the Malthusian Trap: The Paradigm Shift from Input End to Output End

The Core Insight of Malthus

In 1798, Malthus proposed the first rigorous "systemic limit" model in the history of human thought. His insight was not that "population would outstrip food"—that was common sense. His insight was that there exists a structural asymmetry—the gap between the geometric growth potential of population and

the arithmetic growth potential of food—that is not the result of accidental harvest failures or wars, but an inevitable product of the system's internal dynamics. It is this "necessity" that distinguishes Malthus's work from all prior descriptive accounts of famines and disasters. He discovered that "limit" is not an event, but a regularity.

What Complexity Metabolic Failure Inherits

From Malthus, Complexity Metabolic Failure inherits three core legacies.

First, the structural character of limits. The predicament is not "what has gone wrong," but "what constraint the system, under a given architecture, is bound to approach." Malthus located the constraint in the diminishing marginal returns of land; Complexity Metabolic Failure locates the constraint in the boundedness of protocol metabolic bandwidth. The location differs, but the mode of questioning is a direct line of descent: not asking "who made the mistake," but asking "where is the necessary bottleneck of the architecture?"

Second, the dynamic coupling of growth and constraint. Malthus did not say "population growth is a bad thing." What he said was: growth itself will, at a certain point, awaken a constraint that was previously dormant. The dynamic logic of Complexity Metabolic Failure is fully isomorphic: the growth of subsystem interactions themselves—technological progress, financial innovation, information circulation—will, at a certain point, awaken the constraint of protocol metabolic capacity. The problem is not growth; the problem is the mismatch between growth and constraint.

Third, the periodic overshoot-and-collapse as diagnostic signal. What Malthus predicted was not a "one-time collapse," but a cycle of "prosperity—collapse—recovery—renewed prosperity"—the system eternally oscillating around the constraint. The diagnosis of Complexity Metabolic Failure likewise attends to cycles: not "a single crisis," but "the systematic attenuation of old recovery mechanisms." But it locates the end of the cycle not in "more efficient population brakes," but in "architectural upgrading or prolonged congestion."

What Complexity Metabolic Failure Revises

Malthus had two premises, both of which have been falsified by two centuries of empirical fact.

First, he assumed that constraints occur only on the input side—that the chemical productivity of soil is the single unbreakable boundary. The Industrial Revolution proved this assumption wrong: technology can open bottlenecks on the input side—chemical fertilizers broke the land's chemical constraint, fossil energy broke the muscular power constraint, and photovoltaics are now breaking the fossil fuel reserve constraint. Input-side constraints are not absolute; they are functions of technology.

Complexity Metabolic Failure relocates the constraint from the input end to the metabolic end. The constraint on the metabolic end differs fundamentally from that on the input end: input-end constraints can be broken through by substitutes—coal for timber, oil for coal, solar for oil. But metabolic-end constraints cannot be broken through by substitutes—you cannot substitute "faster coordination" for "coordination" itself. Coordination is an act that the system must perform itself; it cannot be delegated to any subsystem. This is why the metabolic-end constraint is "harder" than the input-end constraint.

Second, Malthus assumed that the system's response to the constraint can only be quantitative—famine, war, pestilence, all restoring equilibrium through mortality. This premise has been partly falsified by the spread of contraceptive technology and the establishment of social safety nets. A system can respond to input-end constraints through qualitative adjustments (fertility decline) rather than quantitative catastrophe.

But Complexity Metabolic Failure faces an opposite predicament: for the metabolic-end constraint, the system currently possesses no known "qualitative" automatic regulatory mechanism. When coordination bandwidth is exhausted, the system does not, as it does in the face of famine, spontaneously produce a mechanism for "reducing the number of subsystems"—the survival will of subsystems (firms, institutions, states) drives them to continue interacting, and every interaction further intensifies the congestion. This is why metabolic failure is more dangerous than the classical Malthusian trap: it has no built-in "automatic brake."

Delimitation Summary

Malthus discovered the structural "entrance limit." Complexity Metabolic Failure discovers the structural "exit limit." The former takes the unit of analysis to be material flows; the latter takes it to be information flows and protocol flows. This is not a negation of Malthus, but the extension of his "limit dynamics" research program from the resource domain into the coordination domain.

6.2 With the Limits to Growth: The Paradigm Extension from Material Flows to Information and Protocol Flows

The Core Insight of the Limits to Growth

The 1972 *Limits to Growth* report extended the Malthusian single-resource constraint into the interaction effects of multiple constraints. Population, industrialization, pollution, food production, resource consumption—these five variables were coupled together in a system dynamics model, forming a feedback network. The limit was no longer a line, but a surface: the overshoot of any single variable could trigger cascading effects in others.

This paradigm has profoundly shaped the conceptual foundation of global environmental governance for half a century. Its contribution lies not in predictive precision—the model's specific parameters were long ago outdated—but in bringing "systems thinking" into public policy discourse: what we face is not a scattered list of separate problems, but an interconnected system.

What Complexity Metabolic Failure Inherits

From the Limits to Growth, Complexity Metabolic Failure inherits three core insights.

First, multi-dimensional coupling rather than single-dimensional constraint. The limit is not the shortage of some single element, but the uncontrollable oscillations generated when multiple variables interact. Complexity Metabolic Failure fully internalizes this perspective and takes it a step further in abstraction: it is not a few material variables (population, food, pollution) that are coupling, but all subsystems possessing self-organizing capacity—finance, information, energy, education, military, culture. The dimensionality of coupling expands from five to an arbitrary number—as long as subsystems continue

to generate new interactions, the dimensionality of coupling continues to grow.

Second, the dynamic of overshoot-and-contraction. The Limits to Growth model predicted not a smooth approach, but a sharp contraction after surpassing the carrying threshold. The dynamic logic of Complexity Metabolic Failure likewise contains "overshoot": the system's continued growth within the old architecture (more rules, more technologies, more transactions) will not automatically stop at the metabolic boundary, but will overshoot it—manifesting as functional congestion rather than immediate collapse—and subsequently enter a level of efficacy lower than the threshold.

Third, the systematic character of policy delay. The Limits to Growth model correctly identified that decision-makers' responses to signals involve delays—there exists a long lag between "pollution begins to cause damage" and "policies begin to limit pollution." Complexity Metabolic Failure reconceptualizes this delay as the time lag of protocol metabolic capacity, and elevates it from "insufficient political will" to an inevitable attribute of architecture—any information processing requires time, and delay cannot be infinitely compressed by "faster decision-making."

What Complexity Metabolic Failure Revises

The Limits to Growth thesis has a deep ontological presupposition: constraints ultimately rest on physical variables—population numbers, pollutant concentrations, non-renewable resource stocks. Its modeling takes "material stocks and flows" as the basic unit.

Complexity Metabolic Failure proposes a fundamental revision: expanding the unit of analysis from material flows to information flows and protocol flows.

The justification for this revision is as follows. In the era of the Limits to Growth (the 1970s), the metabolic problems of the system manifested primarily as the material emissions of the industrial economy—sulfur from smokestacks, lead from automobiles, chemical oxygen demand in rivers. Correspondingly, the solutions were also primarily technological substitutes—desulfurization equipment, unleaded gasoline, wastewater treatment plants. These solutions were effective, because the metabolic problems they addressed were those that could be handled independently by peripheral subsystems.

But the metabolic problems of the contemporary era are no longer of this nature. AI ethics, financial algorithmic coupling, the weaponization of supply chains, information ecosystem pollution—these problems cannot be handled independently by any single subsystem, because they are themselves the direct products of the horizontal interactions among subsystems. You cannot solve the AI governance problem with "AI governing AI," nor can you resolve the complexity of the financial system with "more financial instruments." The material footprints of these problems—carbon emissions, energy consumption—are merely their surface layer. Their deeper ontology is protocol-based: the old rules, the old categories, the old narratives can no longer understand and respond to the causal structures of these interactions.

The "material flows" framework of the Limits to Growth cannot subsume these protocol-based metabolic problems. The "information flows and protocol flows" framework of Complexity Metabolic Failure relocates material problems (carbon emissions) as the material facet of a larger metabolic predicament—

not that they are unimportant, but that they cannot be understood and resolved in isolation.

Delimitation Summary

The Limits to Growth extended Malthus's "resource scarcity" into "a systemic bottleneck of multi-variable coupling." Complexity Metabolic Failure extends the Limits to Growth's "material variables" into "information and protocol variables." This is not a rejection of the Limits to Growth, but its incorporation into a more general theoretical framework—a framework in which the physical limits described by the Club of Rome are special cases of Complexity Metabolic Failure in the material dimension.

6.3 With Involution: From Single-Dimensional Margins to Cross-Dimensional Compounding Fallacies

The Core Insight of Involution

The contemporary scholarly usage of "involution" describes the diminishing marginal returns of investment within a single dimension—more labor invested in the same plot of land, raising per-unit yield but lowering per-capita output; more educational credentials invested in the same job market, raising years of schooling but stagnating starting salaries; more time invested in the same competitive track, redoubling effort but leaving rank unchanged. Involution captures the core intuition of "growth without development."

What Complexity Metabolic Failure Inherits

Within the insight of involution are two legacies that cannot be discarded.

First, diminishing marginal returns can occur under conditions of resource abundance. Involution does not require "the pie to have shrunk"; it only requires "the mode of dividing the pie to be such that your additional investment cannot obtain a proportional additional return." This is fundamentally different from the classical scarcity paradigm. Complexity Metabolic Failure fully inherits this insight: functional congestion can likewise occur under conditions of resource abundance and technological advance—not because "the system lacks input," but because "the system cannot convert input into effective output."

Second, exit as a rational signal. Involution theory correctly identifies that when marginal returns diminish below a threshold, the individual's exit—"lying flat," hikikomori, quiet quitting—is not a moral failure, but a rational adaptation to a structural constraint. Complexity Metabolic Failure elevates this logic from individual behavior to systemic behavior: when the net marginal return of the system turns negative (Criterion A), exit occurs not only at the individual level but also at the subsystem level—firms refusing to expand investment, states refusing to bear the costs of public goods, citizens refusing to participate in traditional politics—these are all system-level forms of "lying flat."

What Complexity Metabolic Failure Revises

The fundamental limitation of involution lies in this: it assumes the constraint is dimension-bound—it always occurs within a known, homogeneous competitive dimension. Philip Huang's agricultural involution occurs within the dimension of "land"; Xiang Biao's credential involution occurs within the dimension of "educational investment"; the overtime involution of the internet industry occurs within the dimension of "working hours."

But the contemporary predicament has a feature that the concept of involution cannot accommodate: the most severe damage occurs not "within a single dimension," but at the intersections between different dimensions. The optimal strategy of the environmental dimension (biofuels) damaged the stability of the food dimension (the 2008 food riots); the risk controls of the financial dimension (regulatory compliance) intensified the concentration of the banking dimension (too big to fail); the efficiency improvements of the information dimension (algorithmic recommendation) undermined the foundations of the consensus dimension (post-truth politics).

This is not involution. Involution is the "inward collapse" of homogeneous competition; these are compounding fallacies across heterogeneous dimensions—each dimension optimizing according to its own rationality, but the interactions between dimensions coordinated by no one, the aggregate outcome betraying the original intention of each optimizer.

Complexity Metabolic Failure provides a framework that subsumes both phenomena: involution is the manifestation of metabolic failure within a single dimension; compounding fallacies are its manifestation at the intersections between dimensions. The common etiology of both is the insufficiency of protocol metabolic capacity—when the coordinator cannot synchronously understand and respond to the interactive consequences of multiple dimensions, each dimension can only optimize internally (involution), while the conflicts between dimensions are left unattended (compounding fallacy).

Delimitation Summary

Involution is the projection of metabolic failure in single-dimensional space. Complexity Metabolic Failure retains involution as a special case while incorporating into its subsumptive range the cross-dimensional compounding fallacies that involution cannot accommodate. This extension elevates the unit of analysis from "the margin within a single dimension" to "the protocol across multiple interacting dimensions."

6.4 With Systemic Risk: From Contagion to Autogenic Congestion

The Core Insight of Systemic Risk

Systemic risk theory—an analytical framework that gained wide academic and policy influence after the 2008 financial crisis—describes how, in a tightly coupled network, the failure of a single node can propagate to the entire system through cascading effects. Its classic metaphor is the "domino": Lehman Brothers falls, and then the global financial system is struck in sequence. It rewrote the philosophical foundation of financial regulation: from "supervising whether individual institutions are sound" to "supervising whether the system as a whole is fragile."

What Complexity Metabolic Failure Inherits

The following insights of systemic risk are fully absorbed by Complexity Metabolic Failure.

First, coupling is the latent hazard. Connections among subsystems are not neutral. They can transmit efficiency, but they can also transmit disaster. Complexity Metabolic Failure fully accepts this premise and generalizes it: coupling not only transmits risk, but also continuously generates new information that needs to be coordinated. Risk is only the extreme manifestation of interaction complexity, not its only

form.

Second, micro-rationality sums to macro-irrationality. Each bank is doing prudent risk management, yet their sum produces systemic fragility. The structure of this "compounding fallacy" is the core insight of systemic risk theory. Complexity Metabolic Failure abstracts it further: it is not the sum of "risk management," but the sum of any subsystem optimizing behaviors—environmental policies, technological advances, information dissemination—that can, under conditions of insufficient protocol metabolic capacity, produce compounding fallacies.

What Complexity Metabolic Failure Revises

Systemic risk theory describes an event-triggered dynamic: there is an identifiable triggering event (Lehman's collapse), a contagion chain (counterparty risk exposure), and a terminal stage (financial crisis erupts or central bank intervention succeeds). Its problem frame is: "Where were we hit from, and how do we prevent the next point of impact?"

But in the contemporary predicament, we observe a state that systemic risk theory cannot accommodate: sustained smoldering without a triggering event. No new Lehman Brothers has failed, yet the liquidity illusion in the financial system continues to deepen. No new hot war has been declared, yet globalization is quietly unraveling. No authority has proclaimed "truth is dead," yet the riverbed of consensus continues to dry up.

This state is not "contagion"—because there is no source of contagion. It is autogenic congestion—not the failure of some node being transmitted to other nodes, but all nodes operating normally (or even optimizing), yet the sum of their normal operations already exceeds the information traffic that the protocol can process. Congestion does not need a "bad node" to initiate it. When a road network is saturated, any car driving normally is part of the congestion.

Complexity Metabolic Failure replaces "contagion" with "throughput" as its core metaphor. The contagion model asks: where is the fault? The congestion model asks: how much bandwidth is left? Contagion requires tracing an origin; congestion only requires measuring the ratio of flow to capacity. This explains why, in the absence of a triggering financial event of 2008 magnitude, the sustained accumulation of fragility in the global financial system remains intelligible—it is not waiting for the next contagion, but steadily approaching its protocol saturation bandwidth in the course of normal interactions.

Delimitation Summary

Systemic risk is the manifestation of metabolic failure in the "contagion" form—when the tension of $C > P$ is released in concentrated fashion through a triggering event, it presents as cascading collapse. Untriggered smoldering is the manifestation of metabolic failure in the "chronic" form—when the tension of $C > P$ accumulates continuously but never finds a single release point, it presents as the slow congestion of efficacy. Complexity Metabolic Failure unifies both forms under the same dynamic inequality: both are the result of $C > P$, differing only in whether the release of tension is concentrated or diffuse.

6.5 Summary Table of Delimitations

The table below condenses the core conclusions of the four dialogues of this chapter into a single comparative matrix, allowing the reader to rapidly locate the ecological niche of Complexity Metabolic Failure within the theoretical lineage.

Theoretical Paradigm	Unit of Analysis	Bottleneck Location	Core Logic	Phase-Transition Exit
Malthusian Trap	The dynamic ratio of population to food	Input end (land productivity)	Geometric growth vs. arithmetic growth → famine as brake	Technology opens input-end substitute bottlenecks
Limits to Growth	The compound coupling of population, resources, and pollution	Input–output bidirectional (but ultimate constraint in physical stocks)	Overshoot-and-collapse → multi-variable cascading breakdown	Zero-growth steady state or technological decoupling
Involution	Marginal returns within a single-dimensional competitive field	Within a single dimension (the carrying capacity of one track)	Additional investment → diminishing marginal output → growth without development	Dimensional leap (elevating competition or exit)
Systemic Risk	The failure and contagion of nodes in a coupled network	The transmission channels among nodes	Trigger-cascade → systemic collapse	De-coupling, firewalls, lender of last resort
Complexity Metabolic Failure	The ratio of total information increment from horizontal subsystem interactions to the metabolic bandwidth of top-level protocols	Metabolic end (the throughput bottleneck of information and protocol flows)	$C > P$ → irreversible functional congestion → systematic attenuation of old recovery mechanisms	Architectural upgrading: from collective mind to synergistic cognitive network

Table Note: The Malthusian trap and the Limits to Growth share a bottleneck location on the "input end / material flows." Involution and systemic risk each capture, respectively, special manifestations of metabolic failure in single-dimensional space and in the contagion form. Complexity Metabolic Failure does not replace these theories, but unifies them under a single dynamic postulate ($C > P \rightarrow$ functional congestion), while preserving their respective instrumental value at specific analytical levels.

6.6 The Consolidation and Renewal of a Theoretical Legacy

Neither a wholesale negation of the old categories, nor a simple collage of them. The ecological niche that Complexity Metabolic Failure occupies is that empty center on the theoretical map—surrounded by the old categories, but never before independently named.

Malthus inaugurated the research program of systemic limits, but he located the constraint on the input end, and his successors (the Limits to Growth) extended the constraint into multi-dimensional material flows—both saw bottlenecks on the resource side.

The concept of involution discovered the crucial signal of "growth without development"; systemic risk theory grasped the structural principle that "coupling is the latent hazard"—but the former remained trapped in single-dimensional space, and the latter explained only the contagion form.

Complexity Metabolic Failure unifies these local insights, scattered across different regions of the theoretical map, under a deeper dynamic postulate: the boundedness of coordination bandwidth. It demonstrates that the true location of systemic limits lies not in land, not in energy, not in population, not in any material dimension—but in the system's capacity to understand and coordinate itself.

This is not the "grand unification" ambition of theory. This is a contraction that naturally occurs at the boundaries of theoretical evolution—when multiple categories simultaneously fail in explaining the contemporary predicament, the new category that can infer the common cause from their patterns of failure acquires a logical priority.

It inherits the full legacy of this lineage, while pivoting the arrow from the input end to the metabolic end. The significance of this pivot will be unfolded in Chapter 7—Theoretical Implications and Subsequent Roadmaps.

Chapter Summary

Chapter 6 has completed a systematic dialogue between Complexity Metabolic Failure and four existing theoretical paradigms.

The dialogue with Malthus established the paradigm shift from "input end" to "metabolic end." The dialogue with the Limits to Growth accomplished the expansion of the unit of analysis from "material variables" to "information and protocol variables." The dialogue with involution revealed the dual manifestation of metabolic failure in single-dimensional space and cross-dimensional space—involution is its projection; compounding fallacy is its full face. The dialogue with systemic risk distinguished between the contagion-type disorder and autogenic congestion—systemic risk is the event-release form of metabolic failure; untriggered smoldering is its chronic accumulation.

The summary table of delimitations compressed the four dialogues into a single comparative matrix. This new concept does not replace the old categories, but subsumes them under a deeper dynamic postulate, preserving the instrumental value of each old category at its specific analytical level.

The concept has now been constructed, its boundaries demarcated, its criteria specified, and its theoretical

lineage anchored. The final step is discussion: once this concept is established, what does it imply for our practice? Toward what future research does it point? This is the task of Chapter 7.

Chapter 7: Theoretical Implications and Subsequent Roadmaps

A concept, once constructed, tested, and delimited, must still answer one final question: if this concept holds, what does it imply for how we understand the world and how we act?

This is not a demand that the concept provide solutions—a concept is not an action manual. But a concept does have the obligation to indicate: what previously invisible structures has it rendered visible? What previously misdiagnosed predicaments has it restored to their correct diagnostic category? Toward what direction of subsequent inquiry does it point?

This chapter addresses these questions. Section 7.1 unfolds several core layers of the concept's implications. Section 7.2 clarifies the boundary of this paper—diagnostics, not a prescription pad. Section 7.3 lays down three tracks pointing toward subsequent research, the most important of which points toward the Synergistic Cognitive Network.

7.1 Core Implications: If a Civilization Begins to Metabolically Fail

7.1.1 The Redefinition of "Progress": From Throughput to Metabolic Ratio

If Complexity Metabolic Failure is a structural civilizational constraint, then it compels a fundamental re-examination of the very concept of "progress."

Since the industrial era, the core metric of "progress" has been throughput—more energy, more output, more information, more connections. Progress meant *more*. This standard was reasonable in an era dominated by input-end bottlenecks: when the problem was "not enough," "more" was the solution.

But when the bottleneck shifts from the input end to the metabolic end, "more" is no longer the solution—under certain conditions, "more" is precisely the problem. A system whose protocol metabolic capacity has already reached its ceiling, if it continues to increase input—more regulations, more financial instruments, more information circulation—is not helping the system, but accelerating its functional congestion. This is like continuing to deliver mail to an inbox that is already full, and then attributing the recipient's non-reply to "he isn't trying hard enough."

This does not mean that "growth" should stop. "Stopping growth" is equally unrealistic within the framework of metabolic failure—subsystems continue to operate, they continue to generate interactions, and interactions continue to accumulate information. Complexity will not stop growing because you say "enough," just as the Second Law of Thermodynamics will not cease to hold because you hang up a sign saying "No Entropy Increase Allowed."

The real implication is this: the metric of progress must pivot from throughput to metabolic ratio. The ultimate indicator of a civilization's advancement is not how much it can produce per second, but how much of what it has already produced—the interactive consequences among these products—it can understand and coordinate per second. In rough formulaic form, this pivot can be expressed:

Old concept of progress: $\text{Progress} \equiv \text{Total Output} / \text{Unit Time}$

New concept of progress: $\text{Progress}^* \equiv \text{Protocol Metabolic Capacity} / \text{Interactive Complexity Output}$

When the ratio exceeds 1, the system is learning and adapting. When the ratio equals 1, the system is maintaining—still running, but no longer advancing. When the ratio falls below 1, the system is failing—the faster it runs, the further it moves from health. This redefinition transforms "sustainability" from an ethical slogan into an operationalizable dynamic criterion.

7.1.2 "Why Good Policies Compound into Bad Outcomes" Now Has a Unified Explanatory Framework

In public discourse, a puzzling phenomenon recurs: each individual policy appears reasonable on its own, yet their aggregate effect is disappointing or even harmful. This phenomenon is described by different disciplines using different terms—compounding fallacy, policy offset, unintended consequences, multi-sectoral coordination failure.

Complexity Metabolic Failure provides a unified causal framework for these scattered descriptions.

Various policies are formulated and implemented by different functional departments. Each department optimizes within its own analytical framework. When the protocol metabolic capacity of the system as a whole is sufficient, the cross-departmental coordination mechanism can identify and buffer the unexpected conflicts among policies—an environmental policy may push up food prices, but the coordination mechanism is able to issue a warning and adjust the policy mix before food prices rise significantly.

But when protocol metabolic capacity is persistently surpassed, the bandwidth for such cross-departmental coordination is no longer enough. It is not that the coordination mechanism "doesn't work"; it is that it can only process $N - M$ of the N simultaneously occurring interaction conflicts, and those M unprocessed conflicts become the source of "unintended consequences." As M continues to increase, more and more "unintended consequences" are deposited into the system, forming a sedimented layer of social experience.

Within this framework, "good policies compounding into bad outcomes" is not accidental policy error. It is a systematic product of metabolic failure: when the optimization speed of each subsystem exceeds the metabolic speed of global coordination, the sum of local rationalities necessarily deviates from global rationality. There is no need for anyone to be a bad person; no policy design need be flawed. One single inequality suffices: $\text{Interactive Complexity Output} > \text{Protocol Metabolic Capacity}$.

The practical implication of this explanatory framework is this: it shifts attention from "optimizing each individual policy" to "enhancing the protocol metabolic capacity of cross-departmental coordination." If metabolic capacity is insufficient, ten individually optimized policies, when summed, may yield a total performance inferior to that of three moderately optimized policies that, metabolically, retain enough surplus to coordinate with one another.

7.1.3 Why This Particular Predicament Cannot Be Resolved by "Waiting for the Cycle to Bottom Out"

Cycle theory has held true throughout human history. Prosperity is followed by recession, recession by recovery. Old dynasties fall, new dynasties rise. Old technological paradigms deplete, new technological paradigms erupt.

This cyclical rhythm has made "waiting" a rational strategy: if you believe that the spring is merely compressed, not fatigued, then the optimal strategy is to conserve strength and wait for the rebound.

The framework of Complexity Metabolic Failure points out that this round of predicament may involve not the compression of a spring, but the fatigue of a spring.

Spring fatigue is a precise concept in engineering: when a metal has been repeatedly compressed and released beyond a certain number of cycles, its crystal lattice structure undergoes an irreversible change. It can no longer fully rebound to its original shape. Macroscopically, it can still be compressed and can still release—it appears still to be "working"—but its elastic coefficient has permanently changed.

In the context of civilizational systems, "spring fatigue" means: the recovery mechanisms within the old architecture—the guidance of price signals, the resource reallocation of hegemonic transition, the Schumpeterian creative destruction of technological revolution—still exist, still operate, but their capacity to restore systemic efficacy has permanently attenuated. This is not because the recovery mechanisms themselves are broken, but because the interaction complexity they must process has already exceeded their design bandwidth.

If this is correct, then "waiting for the cycle to bottom out" is no longer a rational strategy. Waiting for an already fatigued spring to fully rebound is not an engineering problem whose variable is time, but a design problem whose variable is architecture: you need to replace the spring, not wait for it.

This is not a prediction that "there will be no recovery this time"—local recoveries will of course occur. But the peak of each recovery may be systematically lower than the peak of the previous one; the trough of each crisis may be systematically deeper than the trough of the previous one. This descending envelope is the macro-trace of metabolic failure on the scale of historical time.

7.1.4 Transforming the "Synergistic Cognitive Network" from a Vision into a Diagnostic Concept

Within the author's theoretical system, the "Synergistic Cognitive Network" is a constructive concept—it describes an idealized civilizational architecture in which the horizontal interactions of subsystems can be perceived and coordinated in real time through a metacognitive interface, endowing the system as a whole with the capacity to observe and regulate its own operation. This concept is anchored in Theorem A0 of *Noospheric Ecology* and further developed in *Civilization Dynamics* as the architectural blueprint for civilizational phase transition.

Within the argumentative framework of this paper, the Synergistic Cognitive Network performs a more foundational function: it serves as the contrastive counterpart to metabolic failure.

The very definition of metabolic failure implies its opposite. If metabolic failure is "Protocol Metabolic Capacity < Interactive Complexity Output," then the healthy state is "Protocol Metabolic Capacity >

Interactive Complexity Output." The Synergistic Cognitive Network is precisely the architectural form that a system must take to maintain this "greater than" state.

This is not to say that the Synergistic Cognitive Network is "the solution to the problem"—that is the task of subsequent work. What this paper aims to establish is this: the Synergistic Cognitive Network is the conceptual dual of Complexity Metabolic Failure. With this dual relation in place, the concept of metabolic failure is no longer an isolated diagnosis, but forms, together with a clearly contoured contrastive state, a complete diagnostic-prognostic axis.

This theoretical posture has an important corollary: it endows the concept of the Synergistic Cognitive Network—previously dismissed as "utopian"—with its diagnostic foundation. The Network is not a beautiful fantasy; it is the necessary answer derived from the question "If metabolic failure holds, what is the only architectural form that can structurally overcome it?" The Synergistic Cognitive Network is the logical response to the proposition " $C > P \rightarrow \text{congestion}$." If the proposition holds, then the only way to structurally reverse the direction of the inequality is not to reduce C (which would require eliminating the self-organizing property of the system—physically equivalent to terminating civilization), but to enhance P (upgrading the protocol metabolic architecture). The form of the Network is thus compelled into shape by the diagnosis itself.

7.2 The Boundary of This Paper: Diagnostics, Not a Prescription Pad

A responsible piece of concept construction must clearly state what it does not do.

This paper does not provide an operational plan for "how to resolve Complexity Metabolic Failure." This is not because the author believes there is no solution—in *Civilization Dynamics*, the architectural blueprint of the Synergistic Cognitive Network, the institutional design of the metacognitive interface, and the conditions and stages of civilizational phase transition have all been given an initial theoretical framework. But the full unfolding of that set of solutions requires the complete argumentative space of another paper.

The function of this paper is diagnostics. Its task is threefold.

First, naming. To bestow a precise categorical term—Complexity Metabolic Failure—upon a syndrome that has long been misdiagnosed.

Second, criteria. To provide draft observational criteria for "how to know when it is happening"—Reversal of Returns, Time-Lag Deadlock, Antagonistic Failure.

Third, direction. To indicate the direction of architectural transformation required to overcome this state—from collective mind to Synergistic Cognitive Network. This direction is not chosen arbitrarily; it is strictly derived from the kernel postulate of metabolic failure: if the problem is insufficient protocol metabolic capacity, then the direction of the solution must be to upgrade the protocol metabolic architecture, not to increase any type of subsystem input.

The prescription is not in this paper, but this paper tells the physician what disease he confronts, and the

basic pharmacological direction of the remedy. The specific prescription is the task of the next piece of work.

7.3 Subsequent Research Roadmaps

This section lays down three of the most direct tracks for subsequent research.

7.3.1 Roadmap One: Operationalization and Empirical Testing of the Criteria

The three criteria proposed in Chapter 5—Reversal of Returns (A), Time-Lag Deadlock (B), Antagonistic Failure (C)—are currently at the stage of draft criteria. The next step is to transform them into operationalizable measurement instruments and to select specific historical or contemporary cases for empirical testing.

Priority empirical directions include:

- Collecting long-time-series data across multiple countries on total word counts of legal texts, the share of compliance costs, and the share of managerial and administrative occupations, to test the cross-system validity of Criterion A;
- Selecting a number of typical "response failure" cases (such as the delay from scientific consensus to international treaty in climate change policy, the regulatory delay for novel financial instruments) and conducting systematic comparative case studies to test Criterion B;
- Using generational survey data and large-language-model-assisted text analysis to track, across multiple societies, the net sentiment scores of core narrative symbols and the concentration of interpretive frameworks, testing Criterion C.

7.3.2 Roadmap Two: The Reinterpretation of Historical Cases

If Complexity Metabolic Failure is a dynamic stage that any complex noospheric system may undergo, then there should be precedents for it in human history. Past historical scholarship has attributed civilizational collapse to single-dimensional factors—corruption, barbarian invasion, climatic catastrophe. The Complexity Metabolic Failure framework offers an alternative interpretive path: to re-examine historical civilizational collapses and test whether they conform to the dynamic signature of metabolic failure—not "the resources ran out," but "the resources were still there, but the system could no longer coordinate its own complexity."

Priority historical case directions include: the administrative bloat of the late Roman Empire, the institutional involution of late imperial China, and the planned coordination failure of the late Soviet Union.

7.3.3 Roadmap Three: From Diagnosis to Design—The Engineering of the Synergistic Cognitive Network

The two subsequent papers in the planned trilogy will directly build upon the diagnostic foundation of this paper.

The next paper—*The Synergistic Cognitive Network: Architectural Conditions for Overcoming Complexity Metabolic Failure*—will complete the turn from "diagnosis" to "design." Taking the draft criteria of Chapter 5 of this paper as design constraints, it will systematically unfold the architectural

principles, core protocols, and transition logic from the old architecture of the Synergistic Cognitive Network. Its relation to this paper is: this paper demonstrates "why and how the old operating system fails"; the next paper demonstrates "what the new operating system should look like."

The paper after that—*The Triggering and Locking of Civilizational Phase Transition*—will return to the phase transition model of *Civilization Dynamics*, fully unfolding the coupling dynamics of the material base *B*, coordination protocols *C*, trust contracts *T*, and meta-narrative *M*, with Complexity Metabolic Failure and the Synergistic Cognitive Network as the two poles of this dynamic—one the critical state before phase transition, the other the locked state after phase transition.

Together, the three papers constitute the "Diagnosis—Design—Engineering" research trilogy of *Civilization Dynamics*. This paper is its first installment, responsible for falsifying old assumptions, proposing a new category, and laying the tracks from concept to measurement.

Chapter Summary

Chapter 7 is the unfolding of implications and the consolidation of directions after the completion of concept construction.

Section 7.1 articulated several core layers of the implications of Complexity Metabolic Failure: it compels the metric of "progress" to pivot from throughput to metabolic ratio; it provides a unified explanatory framework for "why good policies compound into bad outcomes"; it argues that this particular predicament cannot be resolved by "waiting for the cycle to bottom out"; and it anchors its antithetical concept—the Synergistic Cognitive Network—from a vision into a necessary diagnostic dual.

Section 7.2 demarcated the strict boundary of this paper: diagnostics, not a prescription pad. This paper names the disease, provides its criteria, and indicates the pharmacological direction, but does not here unfold the specific prescription.

Section 7.3 laid down three subsequent research roadmaps—the empirical testing of the criteria, the reinterpretation of historical cases, and the engineering leap from diagnosis to design. The position of Complexity Metabolic Failure within this research program is thereby clarified: it is the diagnostic foundation of civilization dynamics.

Next, the final chapter of the paper will restate the core findings and respond to the question that has run throughout: is this concept a death sentence for civilization, or the starting point of its self-awareness?

Chapter 8 Conclusion: In an Age of Spring Fatigue

8.1 Looking Back: Why We Need This Name

The starting point of this paper was an observation: contemporary people are experiencing a peculiar structural aphasia.

We have an ample vocabulary for naming local misfortunes—"involution," "lying flat," "systemic risk," "governance crisis," "climate anxiety," "meaning nihilism." But when we try to articulate the common

grammar underlying these misfortunes, our lexicon abruptly goes blank. Every term captures one piece of the symptoms, but no term can name the basal disease that causes all the symptoms to flare simultaneously.

This aphasia is not a predicament of rhetoric. The impotence of language often signals that a cognitive framework has hit its ceiling. When a civilization cannot name the predicament it faces, it cannot think about it; when it cannot think about it, it can only be driven by it—leaping from one local crisis to another, each time believing "this time is different," each time repeating the same structural exhaustion.

What this paper provides is a name that allows thinking to begin.

Complexity Metabolic Failure. This name describes a dynamic state that can be precisely defined: the information increment generated by the horizontal interactions among subsystems persistently exceeds the perception-interpretation-response bandwidth of the top-level governance protocol. The system does not lack resources, does not lack technology, does not lack effort. What it lacks is the capacity to understand and coordinate itself. And this lack is not a temporary oversight, but an architectural necessity—any complex noospheric system with self-organizing subsystems, upon growing past a certain threshold, will collide with this constraint.

This is not a predicament unique to our era. But it is the first time it has been proposed as an independent category and subjected to the test of a rigorous concept-construction procedure. The Abductive Locking Method has ensured that it is not a clever metaphor, but a theoretical entity tempered by negative delimitation, stress testing, and Occam's Razor. It possesses its own kernel postulate (if the information-generation rate systematically exceeds the information-processing rate, functional congestion necessarily ensues), its own boundary of validity, and its own draft criteria spanning the efficiency layer, the time layer, and the meaning layer.

Now, it is ready.

8.2 This Is Not a Death Sentence

The reader, having come this far, may feel a certain heaviness. If metabolic failure is structural, if its dissolution conditions are not instantiable in human civilization, if we cannot escape by "waiting for the cycle to bottom out"—does this not imply an inevitable, inescapable end?

No. Quite the contrary.

What this paper has argued is not "we are doomed to fail," but "our previous diagnoses of the causes of failure have been systematically misplaced."

Misplacement One: treating a metabolic problem as a resource problem. We thought the problem was insufficiency—not enough energy, not enough technology, not enough growth—and so we kept increasing the input. But the problem is on the metabolic end: the more input, the harder the coordination.

Misplacement Two: treating an architectural problem as a human problem. We thought the problem was

decision-makers who were not smart enough, not moral enough, not strong-willed enough. But the problem is architectural: any decision-maker, regardless of intellectual or moral caliber, is subject to the same protocol metabolic bandwidth ceiling.

Misplacement Three: treating chronic failure as a periodic cold. We thought this was just a more severe recession, and the next recovery would arrive as the last one did. But the spring is not merely compressed—it has grown fatigued.

Correcting these misplacements is not a death sentence. The misplacements themselves were the death sentence—they led civilization, over the past several decades, to repeatedly treat a disease that had not been correctly diagnosed with the wrong medicine, each "treatment" aggravating the condition. Naming the disease is the first step toward ending this vicious cycle.

This is not saying "there is no hope." This is saying: genuine hope lies not in the periodic rebound of the old architecture, but in the clear-eyed recognition of the necessary limits of the old architecture. In human history, every redefinition of systemic limits—from Malthus onward, to the Limits to Growth, and now to the present—was initially misunderstood as pessimism and later retrospectively recognized as a higher form of sobriety. Malthus did not halt the development of agricultural civilization; he merely identified a bottleneck, prompting humanity to search for substitute solutions on the input end. Complexity Metabolic Failure likewise is not an incantation to halt civilizational development; it merely identifies a bottleneck on the output end, prompting humanity to pivot its collective intelligence from "making more" toward "coordinating better."

8.3 The Ultimate Significance of This Concept: Making the Invisible Wall Visible

The greatest contribution that the concept of Complexity Metabolic Failure can make is not prediction, not quantification—those are the tasks of subsequent empirical research—but a more fundamental cognitive act: rendering the "invisible wall" visible.

Public discourse under the old framework has been a debate about the arrangement of furniture in a room without windows. Every crisis triggers a new round of debate—more market or more state? More growth or more distribution? More technology or more regulation? The participants in these debates are all sincere, rational, and well-supported in their arguments. But they all share an unexamined premise: that the room has windows.

What the framework of Complexity Metabolic Failure tries to articulate is this: the problem is not the arrangement of the furniture. The problem is that the walls of this room—the protocol metabolic bandwidth of the current civilizational architecture—are no longer merely a neutral "container"; they have themselves become the bottleneck. Until this wall is seen, until it is understood, any furniture-arrangement scheme—any institutional design, policy optimization, technological substitution—can only achieve local improvements within the bandwidth ceiling of the old architecture. These improvements are real, but they cannot alter the metabolic ratio of the system as a whole. And when a critical mass of people are simultaneously making local improvements—all optimizations competing for the same coordination bandwidth—the local improvements themselves become the source of systemic congestion.

Making the wall visible does not mean we can tear it down immediately. But it does mean: we stop repeatedly crashing into it and then blaming ourselves for not running fast enough.

The Synergistic Cognitive Network has appeared in this paper only as a conceptual dual. It is the logical inverse of metabolic failure—if a system's predicament lies in insufficient protocol metabolic capacity, then the direction of the solution must be to architecturally upgrade protocol metabolic capacity. The specific form of this upgrade is not the task of this paper. But this paper has completed the part of the work that it must complete: providing a precise, testable, and no-longer-obscurable-by-old-categories diagnostic foundation for that new system which must be constructed.

8.4 Final Statement: The Pangs of Civilization's Coming-of-Age

Human civilization is undergoing its coming-of-age.

This is not poetic rhetoric; it is a rigorous statement of system dynamics.

A child civilization knows only how to solve problems on the input end—finding more food, more land, more energy. An adolescent civilization learns, when input-end bottlenecks appear, to seek substitute solutions—fossil energy for timber, chemical fertilizer for fallow land, science for superstition. An adult civilization will confront its own distinct problem, a problem that does not arise at any earlier stage: how to coordinate all that it has already created.

This problem does not need to wait for resource scarcity on the input end before it is posed. It will be posed under conditions of relative input abundance, even surplus—because abundance itself will spawn more subsystems, more interactions, more information. And the consequences of these interactions—from climate to algorithms, from finance to meaning—will not coordinate themselves.

A coming-of-age is always accompanied by pain. What this paper has described—the boomeranging of efficiency, the compounding of policies into poison, the erosion of old narratives, the global synchronicity of exit behaviors—are the forms taken by the pangs of this coming-of-age. The "infinite growth" narrative under the old framework, the "division of labor by department" protocols under the old architecture, were the best tools available to an adolescent civilization. But they are no longer. Not because they have deteriorated, but because the structure of the problem has changed.

The ultimate gift of the concept of Complexity Metabolic Failure lies not in its telling us "we are sick"—many people already feel this. Its gift lies in its providing a rigorous theoretical language that allows us to say, precisely, communicably, and testably: what we are sick with is not some particular disease, but the fact that the architecture itself has become the bottleneck.

This is the cognitive starting point for civilization's transition from the "collective mind"—countless individuals striving in their respective positions, but no one able to see the whole—toward the "Synergistic Cognitive Network"—a civilizational form capable of observing itself, diagnosing itself, and consciously regulating itself.

The starting point is not the answer. But the starting point is the soil in which all answers can grow.

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This reference list follows the principle of "argumentative function labeling": each entry is classified according to the role it actually plays in the argument of this paper. This differs from classification by disciplinary affiliation—a single work may span multiple disciplines, but its function in this paper is singular. The purpose of this classification is to show the reader: what role this reference performs in the argument of this paper.

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Note

This reference list adheres to the following principles.

First, the principle of minimal sufficiency. The argument of this paper does not depend on the authority of any single existing work. The works included in this list are included not because "predecessors have already said similar things," but because they are explicitly cited in the dialogue and delimitation of Chapter 6 or in the draft criteria of Chapter 5, serving as reference coordinates for the theoretical ecological niche. Works not directly cited in the main text are not included in this list.

Second, the priority of the self-sufficient system. Within the theoretical construction of this paper, the author's theoretical trilogy—*Systematic Ontology*, *Noospheric Ecology*, and *Civilization Dynamics*—constitutes the parent system of the concept. These three works are cited multiple times in the main text, but the mode of citation does not presuppose that the reader has fully mastered their contents. The full bibliographic information is provided here to offer a retrieval pathway for readers who wish to trace the deeper roots of the concept.

Third, classified arrangement. The references are arranged in categories according to their functional role in this paper: the author's system as the direct source of the concept, classical theoretical works as the objects of dialogue in Chapter 6, methodological works as the procedural lineage of Chapter 2, and the remainder as the empirical anchors of the draft criteria. This classification is not a division along disciplinary boundaries, but a labeling of argumentative function—it tells the reader what role each reference plays in the argument of this paper.

Fourth, a declaration of openness. This is an initial reference list for a concept-construction paper, not an exhaustive review of the field. As the draft criteria of this paper enter the stage of empirical testing, and as the subsequent papers—on the architectural design of the Synergistic Cognitive Network, on the triggering and locking of civilizational phase transition—unfold, the reference list will expand accordingly. The reference list of this paper is positioned as follows: to provide sufficient coordinate references for the present argument, without claiming to cover all related fields.